Wavence TMN Networking Guide

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Wavence specific

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Basic networking training:

A. Planning and Addressing a Network
B. Basics of IP Addressing
C. Communication in Networks
D. Wavence DHCP Overview
E. Wavence OSPF Overview

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1. TMN Interfaces

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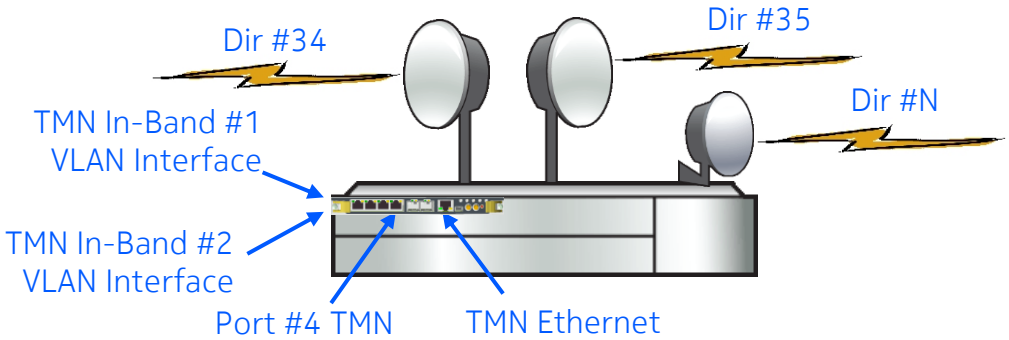
Wavence TMN Networking

Physical Interfaces



Wavence supports a variety of interfaces for TMN traffic:

1. For transport across RF links, there are in-band PPPoE channels, one per Direction
2. The TMN Ethernet port, enabled by default. This interface is intended for local access.
3. User Ethernet Port #4 can be configured for TMN. When provisioned for TMN this interface is intended for TMN connecting to external networks for TMN backhaul.
4. Optionally one or two TMN In-band interfaces. These interfaces can be associated with one or more User Ethernet ports with a user specified VLAN Id. These interfaces are also intended for connecting with external networks. Using these interfaces requires VLAN aware external equipment.
5. Starting with R8.0, TMN in-band is supported on EASv2 card through configuration file. Refer to “Wavence Configuration File Management User Manual”.
6. Starting with R18.0, TMN in-band is supported on EAC card.

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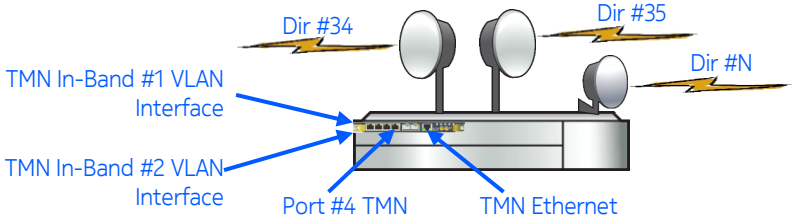
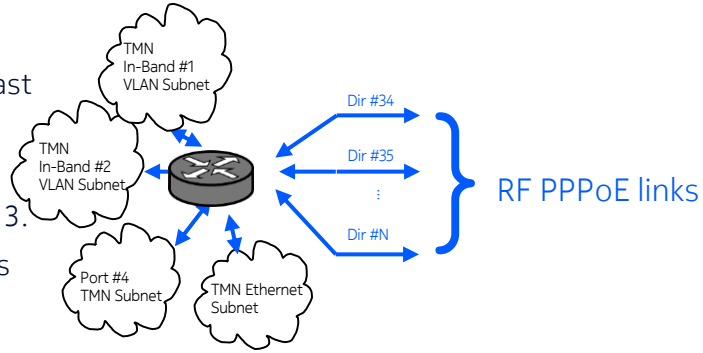
Wavence TMN Networking

Interface to the TMN Router



1. Each TMN interface to the Wavence connects with an internal router.
2. With the exception of the RF PPPoE links, all interfaces to the Router are Broadcast Ethernet interfaces.
3. When these interfaces are provisioned, the subnets must all be unique.
4. TMN traffic passing between any two TMN Network Interfaces is routed at Layer 3.
5. ALL TMN Interface subnets must be unique and cannot overlap. The router does not support Bridging.
6. TMN In-Band VLANs may be associated with one or more User Ethernet ports on the Core, on EasV2 (by config files starting with R8.0) and on EAC (starting with R18.0).
7. All traffic in common TMN VLANs will be switched between the member ports.



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2. TMN Related Services

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Wavence TMN Addressing Support

1. Wavence supports TMN Networking in either IPv4 or IPv6 mode.
    1. Only one mode may be active at a time. Simultaneous support of both IPv4 and IPv6 (Dual Stack) for TMN Networking is not available.
2. Most of this presentation is written using IPv4 examples but the examples apply equally to IPv6.
    1. IPv6 specific behaviors are noted where applicable.

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Wavence TMN Networking

Services - OSPF

Wavence provides two services related to TMN networking.
The first service is OSPF for dynamic routing of TMN network traffic

1. For IPv4 TMN Networks Wavence supports OSPFv2
2. For IPv6 TMN Networks Wavence supports OSPFv3
3. Wavence provides a very basic OSPF implementation
4. User configurable parameters are limited to:
    1. Enabling or disabling OSPF on each individual TMN interface.
    2. Assigning an OSPF Area ID to each interface (default Area is 0)
    3. Enabling or disabling the Stub Flag (indicates whether the interface is a member of an OSPF Stub Area)
5. A single Wavence can function as an Area Border Router (ABR) for up to four OSPF Areas, one of which must be Area 0. A Wavence device may not be a member of more than one OSPF Area unless one or more interfaces are a member of Area 0.
6. Wavence is able to interoperate with external OSPF capable devices such as a 7705 SAR.

Refer to the OSPF Appendix at the back of this presentation for additional information about interoperating with external equipment.

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Wavence TMN Networking

Services – Computer address assignment

In IPv4 mode a trivial DHCP server available only on the TMN Ethernet port.

1. This limited server is intended to support dynamic address configuration of directly connected computers.
2. Enabled by default, the DHCP server can be disabled. This is the only user configurable option.
3. The DHCP server uses an address pool determined by the TMN Local Ethernet IP address and subnet.
4. The maximum number of Addresses managed by the DHCP Server is 10. Clients are served the same Netmask assigned to the TMN Local Ethernet interface and a Default Gateway of the TMN Local Ethernet Port IP address.
5. The Lease Time is fixed to 10 minutes.

Refer to the DHCP Appendix for specifics on how Wavence reserves address for leases.

In IPv6 mode DHCP is not supported. Instead Wavence supports the RFC4861 Neighbor Discovery Protocol (NDP) and sends link-local announcements allowing Stateless Address Auto-configuration of external devices.

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3. TMN IP Addresses

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Wavence Networking

Wavence Addresses

Wavence can be configured with up to five addresses.

The first and primary address is the NE Local Address:

1. This is the Address of the Wavence itself. This is equivalent to the System Address in other types of equipment.
2. <u>The Local Address is the address the WebCT and SNMP Managers must use when monitoring or provisioning the NE.</u>
3. All Wavence SNMP Traps or Notifications are sent from this address
4. All RF PPPoE connections terminating in this shelf (one per Direction) use this address as their PPP Endpoint Identifier.
5. The NE Local Address is reachable by using one of the TMN Interface addresses as a gateway (see the next slide).

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Wavence Networking

Wavence Addresses

Port #4 TMN
Address: 192.168.10.0
Netmask: 255.255.255.192

TMN Ethernet
Address: 172.22.64.86
Netmask: 255.255.255.248

1. The TMN Ethernet interface address
    1. The default interface intended for local access, it is enabled by default.
        1. Caution: Disabling this interface may prevent any direct access to the Wavence. Unless another means of connectivity is available such as Port #4 TMN, TMN In-Band, or an RF-PPP link recovery will require scratching the Wavence Database to return the system to defaults.
        2. Caution: Connecting *this* interface to external networks is possible but not recommended. The default DHCP server may interfere with other external DHCP servers and high volumes of Broadcast or other traffic in the subnet can disrupt Core operations and may trigger unexpected Core restarts.
2. Optional Port #4 TMN Ethernet interface address
    1. This interface is disabled by default. When enabled Core User Ethernet port #4 is reconfigured as a dedicated TMN interface.
    2. Intended for use when connecting to external networks keeping the TMN Ethernet interface available for direct local access.

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Wavence Networking

Wavence Addresses: In-Band interfaces

1. Optional TMN In-Band #1 Ethernet interface address
2. Optional TMN In-Band #2 Ethernet interface address
    1. These interfaces are disabled by default.
    2. When a TMN In-Band interface is enabled a unique VLAN Id is specified for the interface and one or more User Ethernet ports are assigned as members of the VLAN. Traffic in the In-Band VLAN will be switched between all member ports.
    3. These interfaces are intended for connection to external networks
4. Refer to the “Wavence Configuration File Management User Manual” for further configurations.

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Wavence Networking

Wavence Addresses: In-Band interfaces

The primary differences between the In-Band interfaces and the other TMN interfaces are:

1. In-Band TMN Traffic is always VLAN tagged with the user specified VLAN Access to the TMN In-Band traffic requires interfacing with a VLAN aware external device, one that can be configured to use the same VLAN
2. Port flexibility. The user can provision one or more User Ethernet ports that will be used with the In-Band interface
    1. When multiple User Ethernet ports are associated with a TMN In-Band interface the associated TMN VLAN will be switched between the member ports.
3. User ports used for TMN In-Band:
    1. May not be port segregated.
    2. May be associated with PDH cross-connections.
    3. When in 802.1Q mode the ports are always a member of the Wavence internal VLAN 1 and may also be a member of additional User defined VLANs.
    4. When in 802.1ad mode the ports are always a member of the Wavence internal S-VLAN 1 and may also be a member of additional User defined VLANs.

The user should be aware of these behaviors when transporting TMN Traffic in a VLAN through the same physical interface used for external backhaul or when interconnecting User Ethernet traffic between stacked Wavence shelves.

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Wavence Networking

Wavence Addresses – Tips and Limitations

1. Tip: When assigning addresses to TMN Interfaces, the Local Address may be set to match the address of one external TMN Interface but no more than one.
2. The Local Address must match the address of an external TMN interface if it is in the same subnet. *
3. The Wavence Local Address must be reachable from the external network. The Local Address must be reachable from SNMP Management servers, FTP/SFTP servers hosting upgrade software for download, and external NTP servers. If the NE Local Address cannot be reached then some types of SNMP Management functions will fail, some provisioning operations will fail, Remote Software downloads will fail and the Wavence NTP client will fail. A TMN Interface address is not a substitute for the Wavence Local Address.

* See the Known Issues section in the Appendix

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4. The TMN Network

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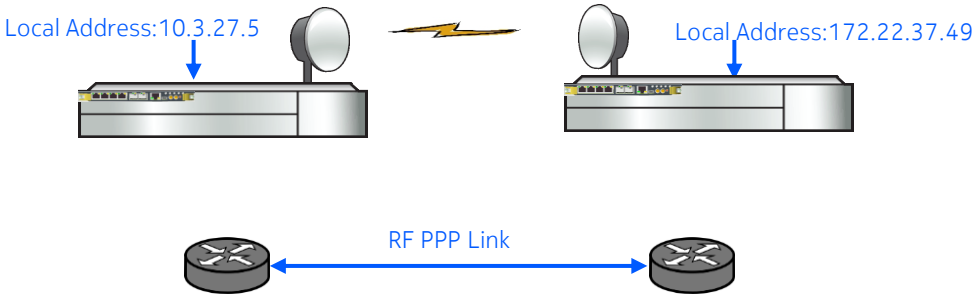
Wavence Networking

The Basic TMN Network – RF PPP Links



1. When enabled, the RF PPP link comes up as soon as the Radio channel is operational.
2. It doesn't matter what Local IP Address is assigned at either end, as long as it is unique, and the IP version is the same (IPv4 or IPv6). When the Radio link is up, the routers can exchange packets with each other.
3. TMN packets exchanged between the routers travel over the RF link in a high priority internal queue.

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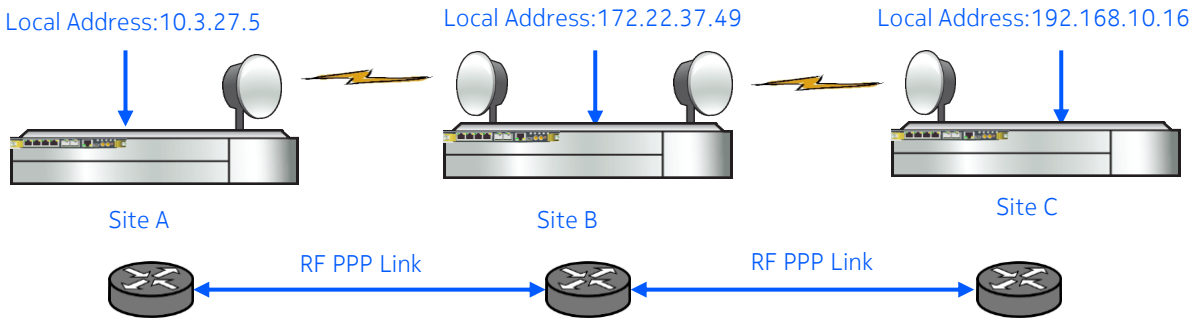
Wavence Networking

The Basic TMN Network – RF PPP Links



1. If we move beyond a single hop, when the RF links are up:
    1. The NE at Site A can communicate with Site B
    2. The NE at Site B can communicate with Site C
    3. but A cannot communicate with C until routing is configured.
    4. For routing to function, the NE addresses must all be unique. If there are any duplicate addresses, or if all addresses are the same (factory default) then routing will fail!
2. Routing can be configured dynamically or statically.
    1. The recommended configuration is to enable OSPF within the Wavence network for dynamic routing.

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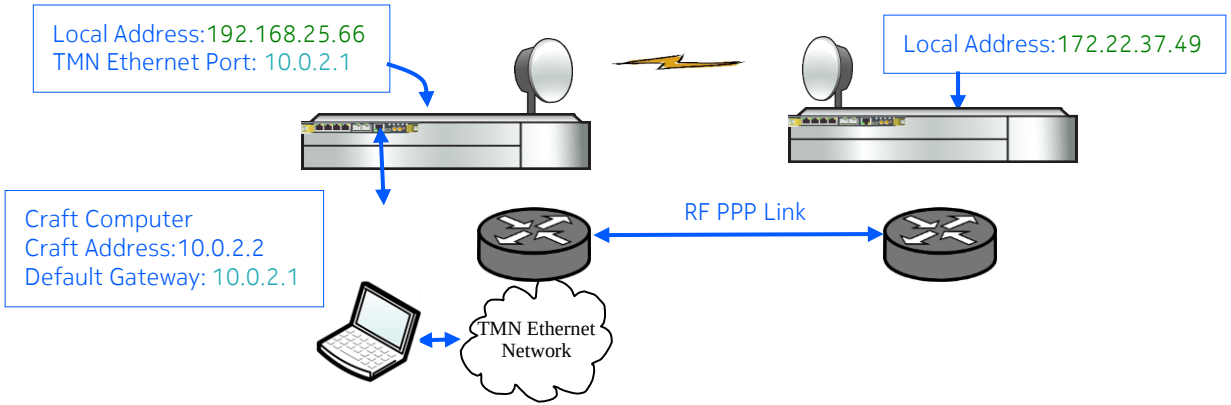
Wavence Networking

The Basic TMN Network



1. Before external devices can gain access to this network, at least one external TMN interface must be provisioned.
2. If we connect a properly addressed Craft computer to one of the TMN Ethernet Interfaces we should be able to communicate with all the Wavences systems in the network, provided the Craft computer is configured to use the **TMN Ethernet port** address as the **Gateway** to the Wavence network and OSPF is enabled or suitable static routes have been provisioned in the Wavence systems.
3. To communicate with either NE specify the Local Address of each Wavence when connecting with the Craft: **192.168.25.66** and **172.22.37.49** in this example.
4. By default the Wavence DHCP service is enabled. If the network interface of the computer is set to “Obtain an IP address automatically”, the computer will be configured correctly when connecting to the TMN Ethernet Port. No user action will be required.
5. When configured for IPv6 Wavence provides link-local announcements using the Neighbor Discovery Protocol (NDP). This allows computers to configure themselves through stateless auto-configuration.

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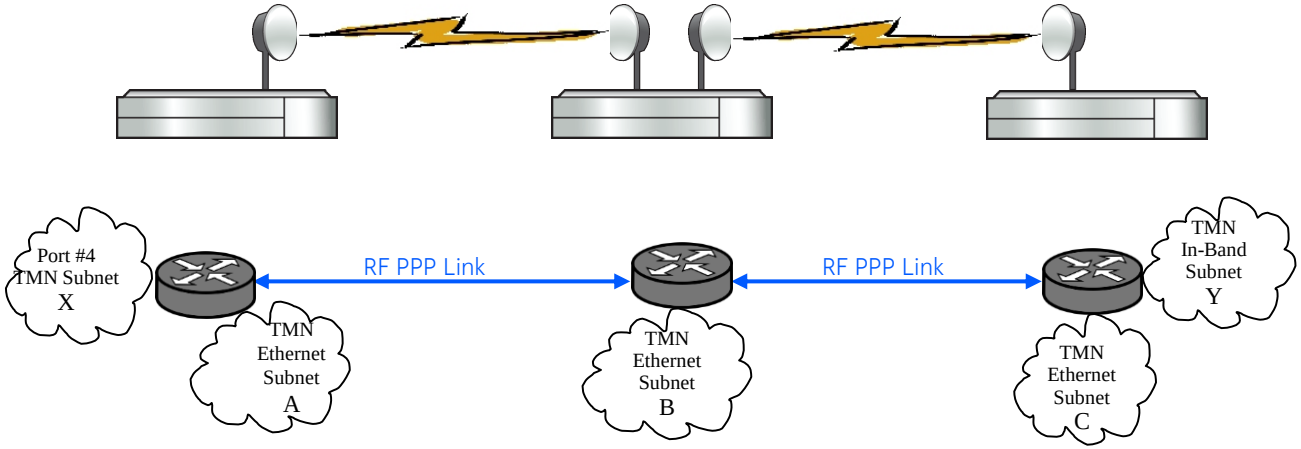
Wavence Networking

A Simple Linear Network



1. To provide local access to each Wavence system and to connect external equipment to the TMN network, we use TMN interfaces at each site. Each interface functions as gateway to the TMN Network.
   1. From a TMN perspective, we have a network of Routers interconnected with PPP links.
2. Each TMN Interface subnet must be unique in the network. Subnets used at one interface cannot be reused at another interface or site within the same Radio network. Subnets on different interfaces must not overlap.
3. All TMN traffic is routed. Bridging between Ethernet subnets is not supported.

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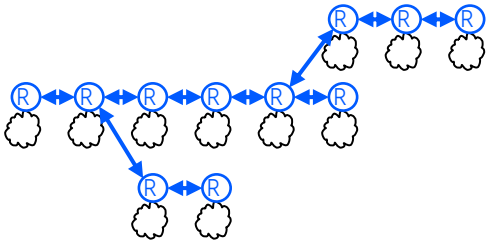
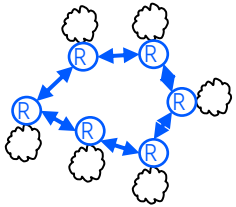
Wavence Networking

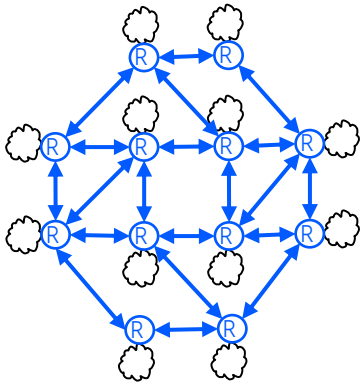
Supported TMN Layer 3 Network Topologies



Linear and Tree



Ring



Mesh

1. The TMN Network operates entirely at Layer 3. This network can be configured in Linear, Tree, Ring, or Mesh topologies.
2. With Ring and Mesh networks, OSPF can dynamically update the routing to take advantage of alternate routes for TMN traffic in the event of a link failure.
3. OSPF will manage routes to prevent Layer 3 loops in the TMN Network only.

Notice: The Ring and Mesh topologies depicted are ONLY applicable for TMN Networking at Layer 3. All Wavence user data transport is performed at Layer 2. It is mandatory that the data transport network be configured to prevent any Layer 2 Ethernet loops through the appropriate provisioning of 802.1Q or 802.1ad VLANs, Ethernet Ring Protection, Port Segregation, or other physical means.

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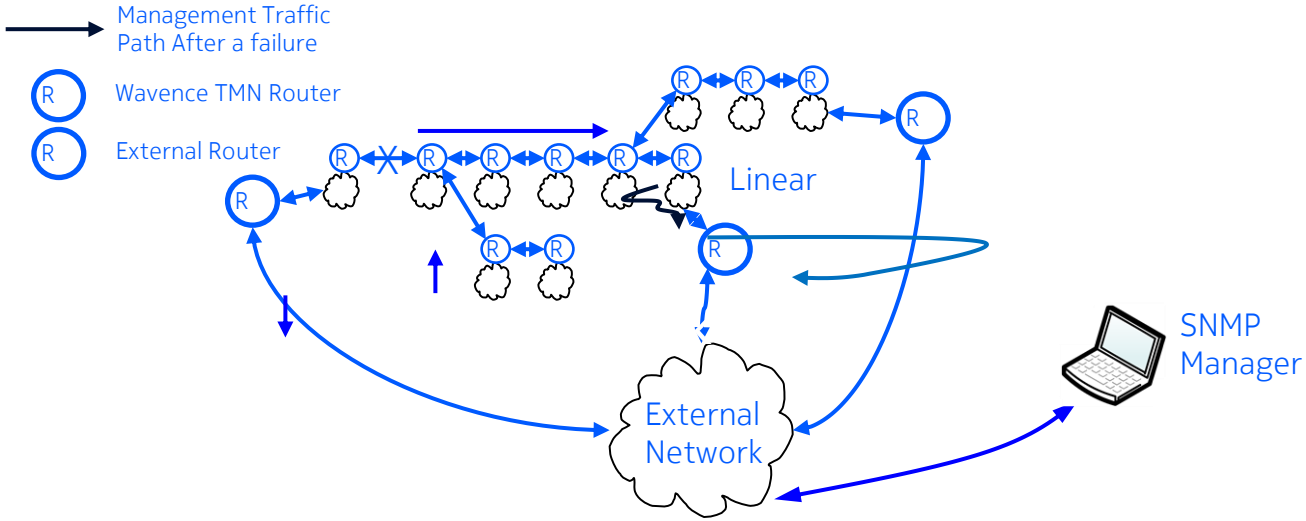
Wavence Networking

Supported TMN Network Topologies



1. Wavence TMN Networks can have multiple external gateways that allow alternate management paths in the event of an outage.
2. To make the best use of multiple gateways, OSPF must be enabled throughout TMN Network.
3. To maximize the availability of alternate TMN network paths, OSPF should be used to manage the links between and any external routers and a suitable dynamic routing protocol should be used in the External Network.

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5. DCN Interconnections examples

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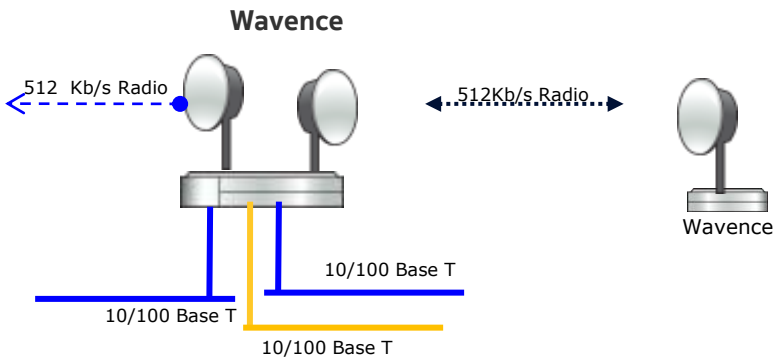
Wavence (MSS-4/8)

Capability IPv4 parameters



Management Protocol:

SNMPv2c/V3 for supervision
FTP/sFTP for download/backup/restore
Telnet/SSH for WT-CLI
NTP for time synchronization
HTTPS for WebCT

Physical DCN Ports:

1x Ethernet RJ45 10/100 BaseT for local access with DHCP (Out of Band)
1x Ethernet RJ45 10/100/1000 BaseT, SW configurable (Out of Band)
Several Ethernet RJ45 10/100/1000 BaseT connections **(In Band)**
Several Ethernet SFP connections **(In Band)**
Nx Radio links 512 Kb/s

IP addressing

1x IP address for all the radio channels (/32) that is the same as the LocalIP address of the NE.
1x IP address and associated net-mask for each TMN Eth port in use

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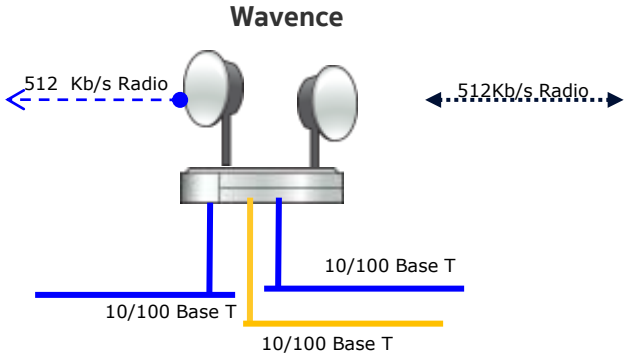
Wavence (MSS-E/HE/XE)

Capability IPv4 parameters



Management Protocol:

SNMPv2c/V3 for supervision
FTP/sFTP for download/backup/restore
Telnet/SSH for WT-CLI
NTP for time synchronization
HTTPS for WebCT

Physical DCN Ports:

1x Ethernet RJ45 10/100 BaseT for local access with DHCP (Out of Band)
Several Ethernet RJ45 10/100/1000 BaseT connections **(In Band)**
Several Ethernet SFP connections **(In Band)**
Nx Radio links 512 Kb/s

IP addressing

1x IP address for all the radio channels (/32) that is the same as the LocalIP address of the NE.
1x IP address and associated net-mask for each TMN Eth port in use

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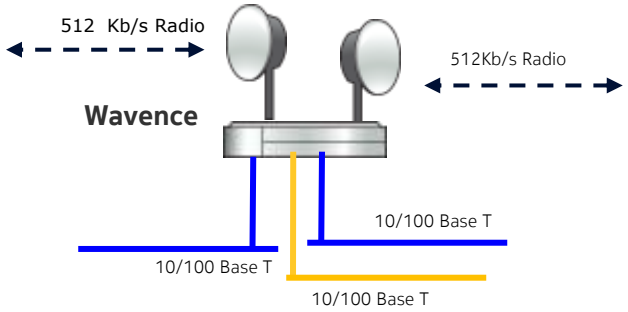
Wavence (MSS-4/E/HE/XE/NIM)

IPv6/OSPFv3 Capability Overview



Overview

1. In line with IPv4 DCN. Some constraints detailed here and in the Known Issue section
2. IPv6 DCN cannot be managed by 1350/1353 NM
3. No support for dual stack IPv4/IPv6 (only one active stack is managed)
4. DHCPv6 is not supported. Instead, IPv6 Stateless Auto-configuration is supported on TMN Local ETH interface

Management Protocols

1. SNMPv2 and SNMPv3 for supervision
2. FTP and SFTP for download/backup/restore
3. Telnet and SSH for Ethernet Features Shell
4. HTTPS for WebCT
5. NTP for time synchronization

Physical DCN Ports

1. 1x Ethernet RJ45 10/100 BaseT for local access without support of DHCPv6 (Out of Band) – IPv6 Stateless Auto-Configuration is supported (always enabled)
2. 1x Ethernet RJ45 10/100/1000 BaseT SW configurable (Out of Band) – IPv6 Stateless Auto-Configuration is supported (always enabled for TMN interface)
3. Several Ethernet RJ45 10/100/1000 BaseT ports (**In Band**)
4. Several Ethernet SFP ports (**In Band**) **Remark** – a maximum of 2 TMN in-band interfaces can be configured for each node
5. nx Radio 512 Kb/s

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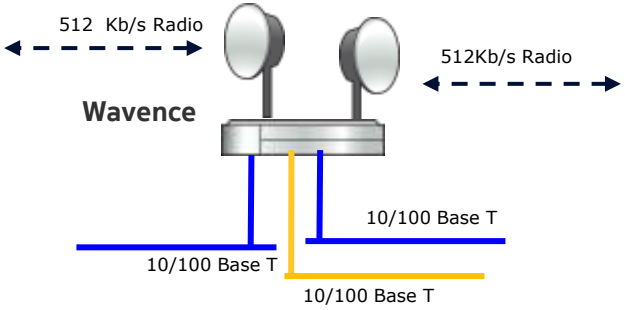
Wavence (MSS-4/8/E/HE/XE/NIM)

IPv6/OSPFv3 Capability Overview



IPv6 routing

1. MSS-4/8/E/HE/XE and NIM all have routing capability. From DCN/IP point of view it is a router.
2. Each port can use OSPFv3 or static routing; for each MSS-4/8/E/HE/XE and NIM up to 3 additional OSPF Areas can be provisioned, plus the backbone.
   1. Recommended deployment is with maximum 2 OSPFv3 areas per node (including backbone)
   2. Remark – OSPFv3 Router ID must be manually provisioned to a unique value for each node (see System Settings) – this is mandatory
3. OSPFv3 areas can be either Totally Stub or “not Stub”.
4. Recommended dimensioning is to limit the maximum number of direct neighbors connected to a Wavence Ethernet interface to 8

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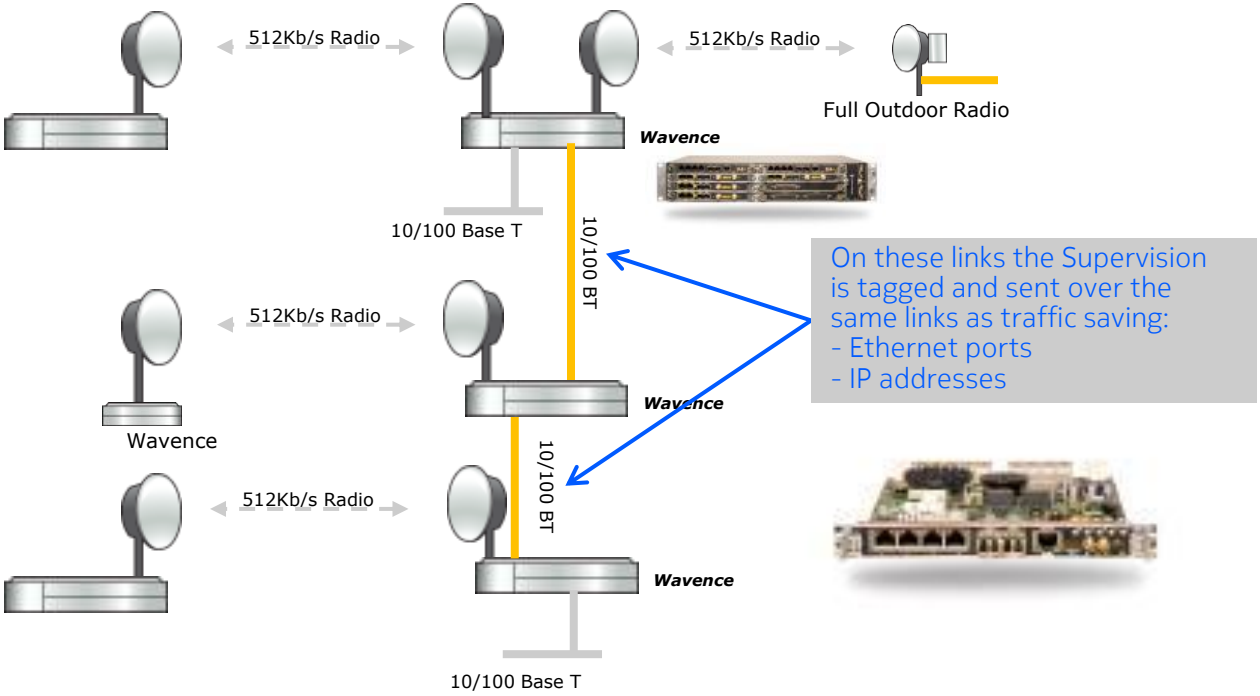
Wavence DCN Interconnection

In Band (InB)



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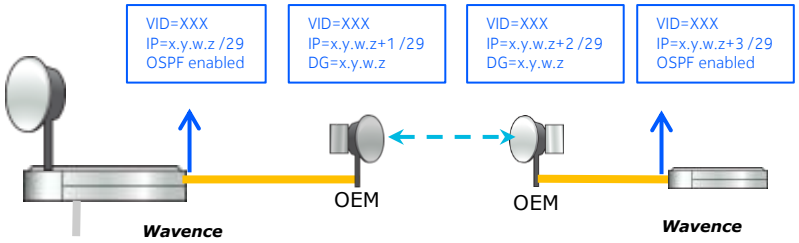
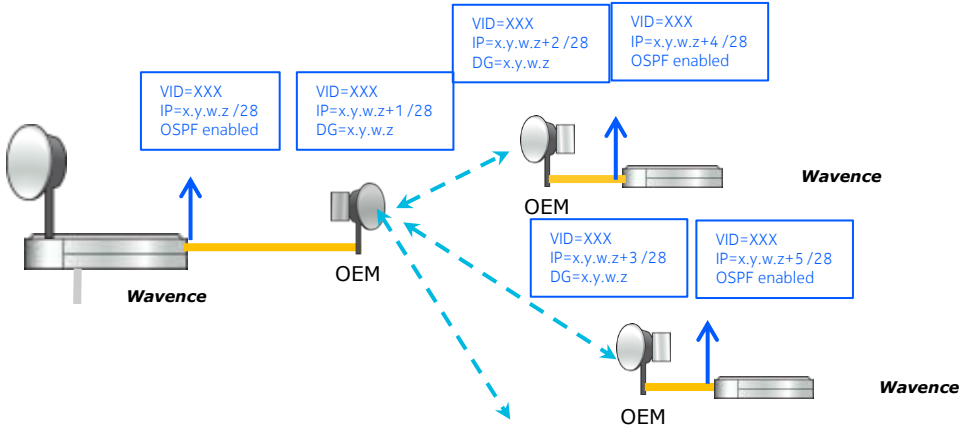
Wavence DCN Interconnection

In Band (InB) w/OEM(MPT-GS, MPT-GM,MPT-SUB6)



OEM
MPT-SUB6 PtP
MPT-GS
MPT-GM



OEM
MPT-SUB6 PtMP

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6. Basic Wavence Address Provisioning

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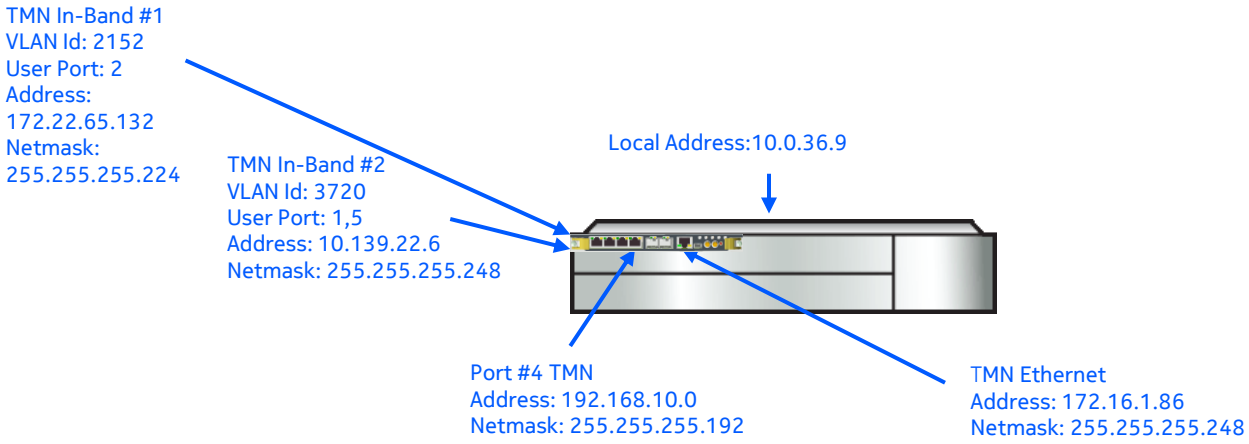
Wavence Networking

Wavence addressing

How many addresses does a Wavence device need?
A better question might be: How much address space does Wavence require?
To answer these questions, we need to know how the interfaces are used in various configurations.



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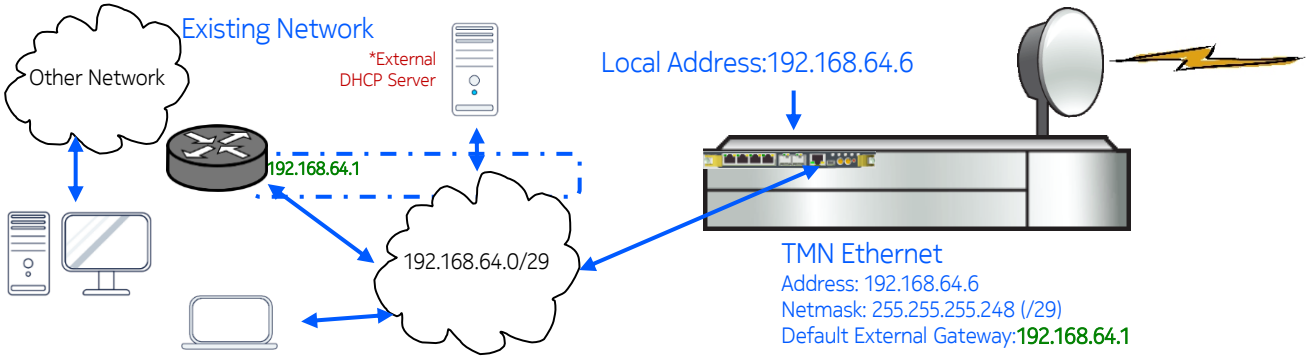
Wavence Networking

Wavence addressing example – Adding a Wavence device to an existing network



1. In the simplest case if Wavence is connected to a *small* existing external network <u>defined by an external Router</u> only requires <u>one</u> IP Address.
2. In this configuration, the TMN Ethernet interface is assigned an address and netmask from the existing subnet, and the Local Address can be set to match. Only one IP Address is needed.
3. Craft computers interface to Wavence via the external network.
4. For external networks larger than about a /29 the TMN Port #4 or one of the TMN In-Band interfaces should be used reserving the TMN Ethernet Port for local access.

*If the router or other external device is providing DHCP service in the subnet the internal DHCP server for the TMN Ethernet Port must be disabled before connecting Wavence to the external network in order to prevent conflicts!

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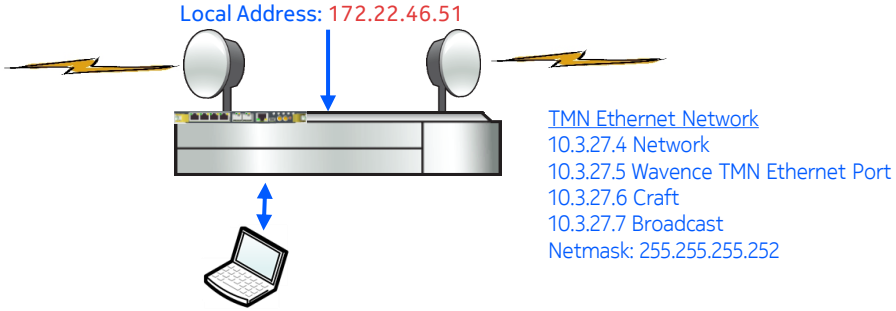
Wavence Networking

Wavence addressing example – Basic Addressing

When Wavence is installed at a location where no pre-existing external network is available, it must at a minimum, provide a network that can be used for local Craft connections.

All TMN user interfaces to Wavence are Ethernet Broadcast interfaces. The smallest network supported is a /30. This size network supports two useable addresses one for the TMN Ethernet port and one for a Craft computer. The Wavence internal DHCP server should normally be enabled (the default) to configure the network interface on the Craft computer when it connects.

In this example the Local Address is set differently from the TMN Ethernet address. Wavence and associated equipment use the space of 5 IP addresses: 4 IP addresses for the TMN Ethernet (/30) network supporting a Craft computer and one for the (different) Local Address.



Remember: To provision or manage this NE, SNMP Manager must specify the 172.22.46.51 Local Address regardless of whether the physical connection is local or remote.

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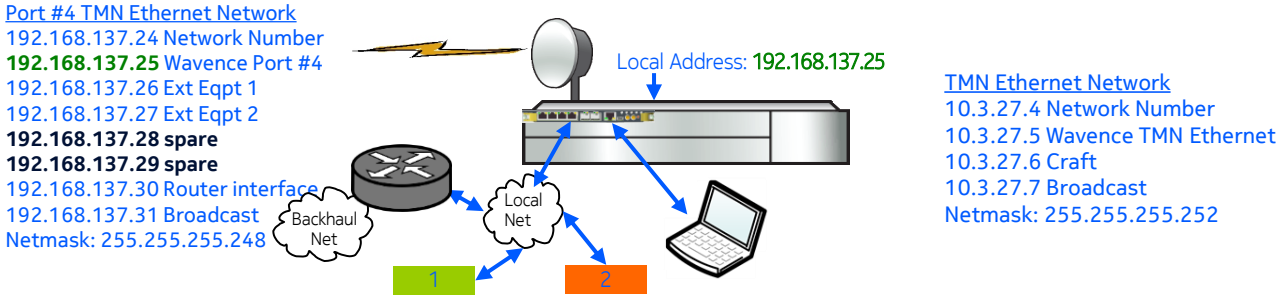
Wavence Networking

Wavence addressing example – Basic Addressing



In this example we have an external router for TMN backhaul a few local external devices and use Port #4 in TMN mode. As shown the smallest subnet that can provide sufficient addresses is a /29. The figure above shows how the addresses could be assigned.

By setting the Local Address and Port #4 address to be the same Wavence will appear to be part of the same subnet as the external equipment from a management perspective.

This configuration requires the space of eight addresses in the Port#4 TMN network, plus the space of four addresses for the TMN Ethernet network for a total space of 12 addresses. Note: The spare addresses left over cannot be deployed at another site. They remains part of this subnet but are available for future site expansion. If the router were the only external device and addresses for external equipment were not needed the router could connect directly to TMN Port #4 and a /30 network could be used.

Be aware that all TMN traffic is transported through the Wavence network at high priority. Normal Wavence TMN traffic is relatively low bandwidth. Although TMN traffic through Wavence is rate limited a high volume of Ethernet traffic to and from external equipment via the Wavence TMN Network may impact revenue bearing traffic. This should be considered during the network design phase.

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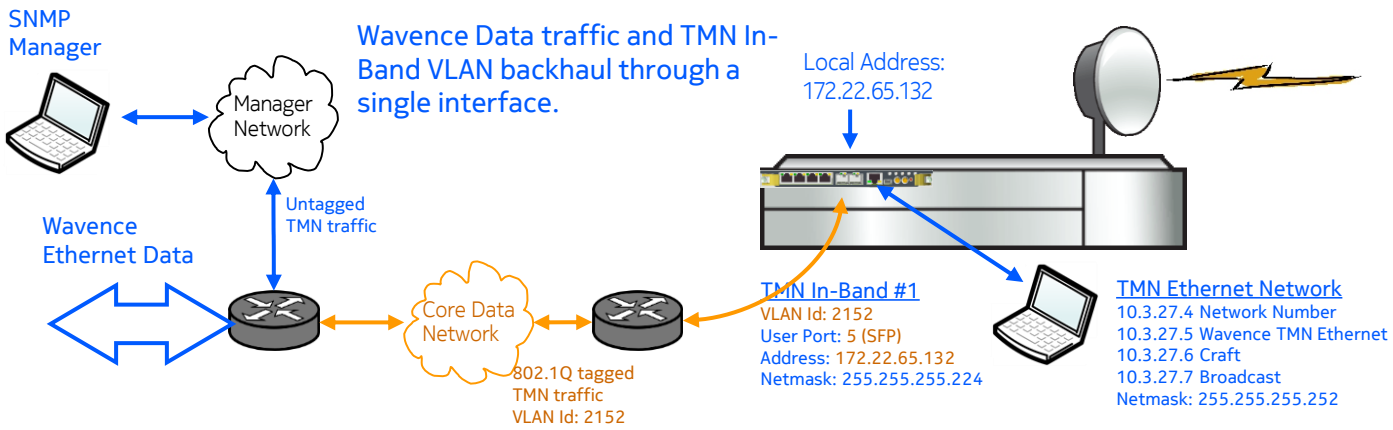
Wavence Networking

Wavence addressing example – TMN In band Interface



1. This example shows TMN In-Band #1 configured to use Optical SFP Port #5
2. Wavence data traffic also flows through this same interface. The TMN In-Band VLAN Id must be unique in the data flow so the TMN traffic can be identified and directed appropriately by the external equipment.
3. Intermediate routers forward the combined TMN In-Band VLAN and Wavence data traffic to appropriate destinations. TMN In-Band traffic can be directed to the appropriate destination using the VLAN Id. In this example the final external router strips the TMN VLAN Id tags from the TMN packets on egress and sends them to the Manager network. Conversely, TMN Traffic coming from the Manager and destined for the TMN Network must be tagged at ingress with the TMN VLAN Id before heading to Wavence.

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7. Interconnecting Multiple Wavence Shelves

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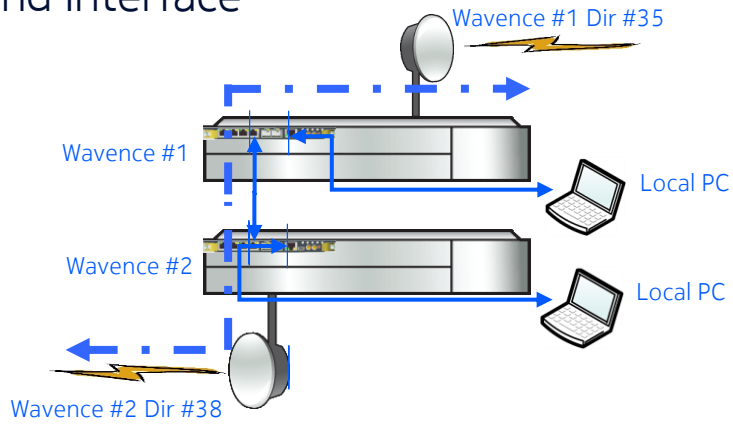
Wavence Networking

Wavence addressing example – TMN In band Interface

Connecting TMN between two Wavence shelves

IP Addr	Equipment	Comment
10.3.27.64	Network	Netmask: 255.255.255.252
10.3.27.65	Wavence #1 Port #4	Also Wavence #1 Local Address
10.3.27.66	Wavence #2 Port #4	Also Wavence #2 Local Address
10.3.27.67	Broadcast	



1. Port #4 TMN ports are addressed in the same /30 subnet and connected together.
2. TMN Traffic flowing through the site via Wavence #1 Dir #35 and Wavence #2 Dir #38 will be routed through the TMN Port #4 interfaces.
3. Any Wavence Data traffic must be cabled separately between shelves. This interconnect method offers maximum flexibility with respect to how data traffic is interconnected between the two shelves.
4. Each TMN Ethernet interface is configured with it's own unique /30 subnet and the DHCP server in each Wavence is enabled (details not shown). This configuration permits independent local access to each Wavence as well as the entire TMN Network.
5. The site consumes the space of 12 addresses, four for the Port #4 /30 and eight more for the two /30 Craft networks.

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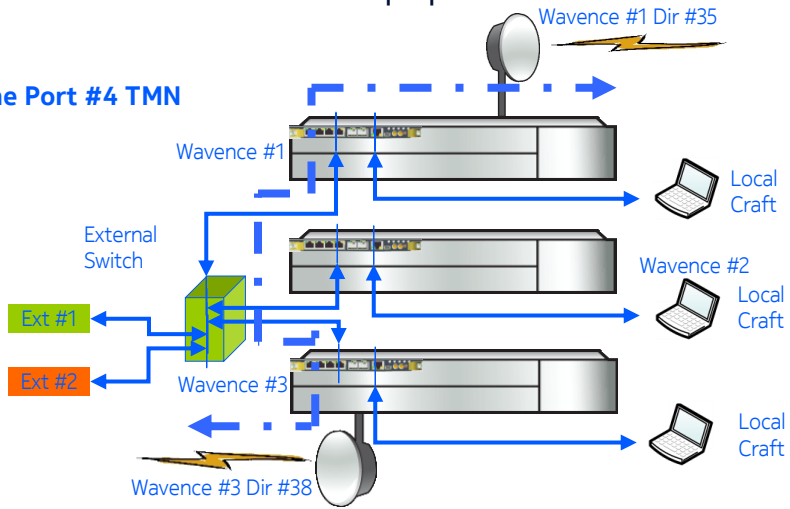
Wavence Networking

Wavence addressing – stacked shelved with external equipment and switch

Connecting multiple Wavence shelves at a site using the Port #4 TMN interfaces

<u>Port #4 TMN Network</u> (/29)
10.3.27.64 Network
10.3.27.65 Wavence #1
10.3.27.66 Wavence #2
10.3.27.67 Wavence #3
10.3.27.68 External Eqpt #1
10.3.27.69 External Epqt #2
10.3.27.70 Spare
10.3.27.71 Broadcast
Netmask: 255.255.255.248



1. All Port #4 TMN ports are addressed in the same /29 subnet and connected together with an external switch. External site equipment may be connected to the TMN Network using via the switch.
2. TMN Traffic flowing through the site via Wavence #1 Dir #35 and Wavence #3 Dir #38 will pass through the external switch. Wavence Data traffic must be cabled separately between shelves.
3. Local management is through the TMN Ethernet interfaces (details not shown).
4. The site consumes 20 addresses, eight for the /29 Port #4 network and twelve for the three /30 Craft networks.

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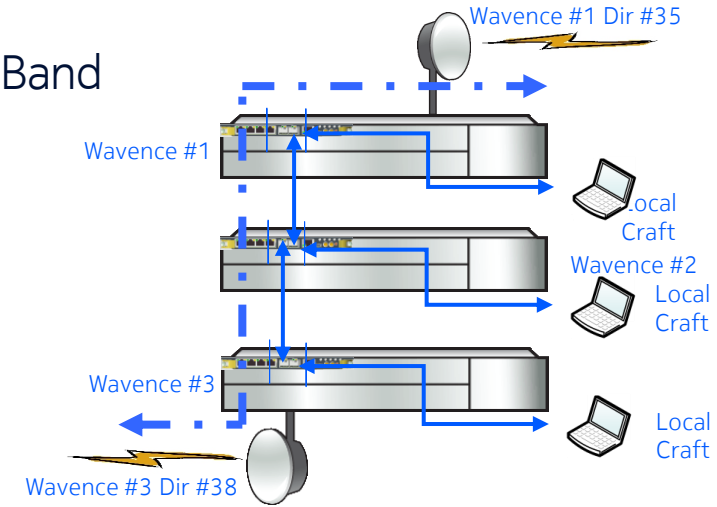
Wavence Networking

Wavence addressing – stacked shelved with TMN in Band

Connecting multiple Wavence shelves at a site using TMN In-Band interfaces

TMN In-Band Network
10.3.27.64 Network
10.3.27.65 Wavence #1, VLAN Id 4079, optical Port #6
10.3.27.66 Wavence #2, VLAN Id 4079, optical Ports #5 and #6
10.3.27.67 Wavence #3, VLAN Id 4079, optical Port #5
10.3.27.68 Spare
10.3.27.69 Spare
10.3.27.70 Spare
10.3.27.71 Broadcast
Netmask: 255.255.255.248

Similar to the previous slide but by using the In-Band Interfaces switch functionality is moved into Wavence #2. Shelf-to-shelf Data traffic is able to share the same cables.



1. All TMN In-Band ports are addressed in the same /29 subnet and use the same VLAN 4079. Putting all In-Band ports in the same subnet avoids an extra routing hop between Wavence #1 and Wavence #3. In larger sites using a single subnet for interconnection will improve address efficiency.
2. Wavence #2 shelf has two User ports associated with In-Band VLAN 4079 and provides the switch functionality to connect Wavence#1 with Wavence #3. TMN Traffic flowing through the site via Wavence #1 Dir #35 and Wavence #3 Dir #38 will pass through the In-Band VLAN 4079 optical switched ports in Wavence #2
3. Shelf-to-shelf Wavence Data traffic can be transported across the same Layer 2 physical connections using separate VLANs. See the notes on In-Band Interfaces in Section 3.
1. Optical ports are shown in this example. A mix of Electrical and Optical or all Electrical are possible. Since Data traffic also uses these connections using 1000Mb/s speed is recommended for Electrical ports.
4. Local management is through the TMN Ethernet interfaces (details not shown).
5. The site consumes the space of 20 addresses, eight for the /29 and twelve for the three /30 Craft networks.

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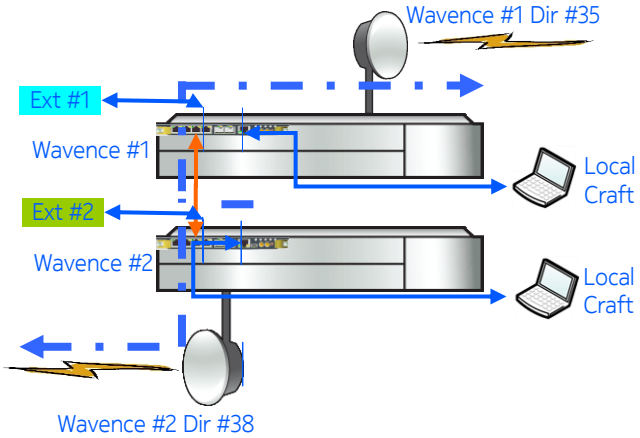
Wavence Networking

Wavence addressing – Another shelf interconnect using TMN in Band

Another variation connecting multiple Wavence shelves at a site using TMN In-Band interfaces, external equipment, and no external switch

IP Addr	Equipment	Comment
10.3.27.64	Network	Netmask: 255.255.255.252
10.3.27.65	Wavence #1 Port #4	Also Wavence #1 Local Addr
10.3.27.66	Ext #1	External Equipment
10.3.27.67	Broadcast	
10.3.27.68	Network	Netmask: 255.255.255.252
10.3.27.69	**Wavence #1 In-Band #1**	**Eth Port #3, Vlan ID: 1001**
10.3.27.70	**Wavence #2 In-band #1**	**Eth Port #3, Vlan ID: 1001**
10.3.27.71	Broadcast	
10.3.27.72	Network	Netmask: 255.255.255.252
10.3.27.73	Wavence #2 Port #4	Also Wavence #2 Local Addr
10.3.27.74	Ext #2	External Equipment
10.3.27.75	Broadcast	



1. This example uses a total of five /30 subnets. This configuration is expensive in terms of the amount of address space required (Space of 20 addresses for 6 devices) but requires no external switch while supporting one external TMN device per Wavence shelf.
2. The two TMN Ethernet ports are dedicated for Craft use with DHCP enabled. Craft addressing is not shown but would be the same as in other examples.
3. The path for TMN Traffic flowing through the site via Dir #35 and Dir #38 is via the TMN In-Band interconnection.
4. Shelf-to-shelf Wavence Data traffic will also flow across the same Layer2 physical connections. See the notes on In-Band interfaces in Section 3.

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8. Management Communication Requirements

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Wavence Networking

Management

We know up to five addresses can be assigned to each Wavence,
Which address must be used for WebCT management?
Which address must be used for SNMP Management?

TMN In-Band Interface #1
Address: 10.78.101.0
?

Local Address:172.22.46.83?

TMN In-Band Interface #2
? Address: 172.22.65.0

Port #4 TMN
Address: 192.168.10.0
?

TMN Ethernet ?
Address: 10.0.36.9

The Local Address is used to provision and manage Wavence.

Connecting to the wrong address is one of the primary reasons for provisioning problems with Wavence!

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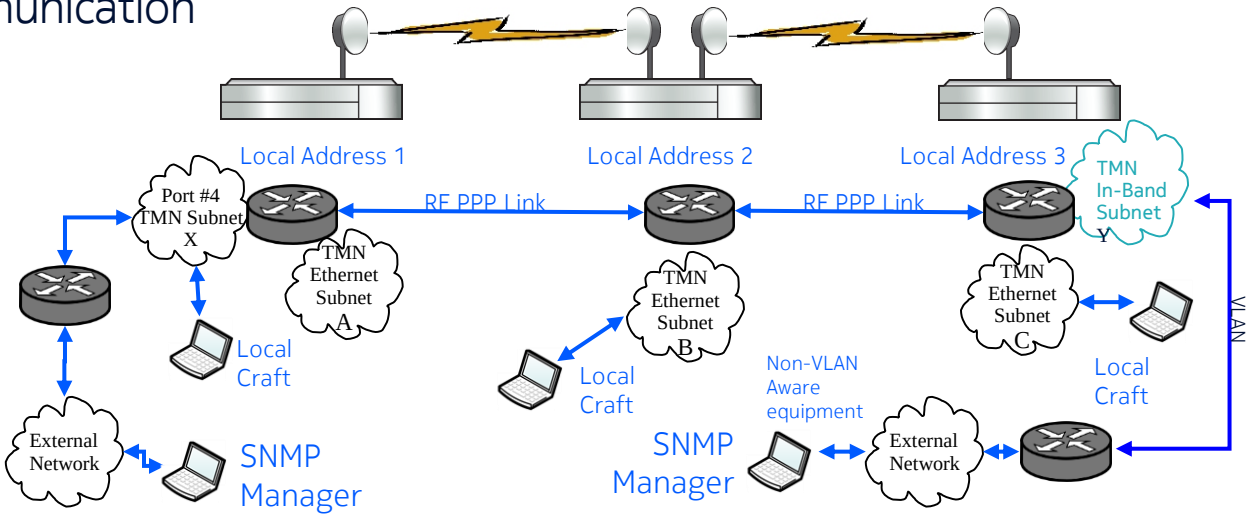
Wavence Networking

Network communication



1. Communication requirements with external networks
2. SNMP packets from the Local Address of each NE must have a route to the managers. This is usually provided by either a Static Default Route at the Wavence Network borders, or learned via OSPF from external Neighboring routers.
3. External routers must be either manually configured to use Wavence as the gateway to the network of Local Addresses or they must learn the gateways to the Wavence network by exchanging routes with Wavence through OSPF.
4. **TMN In-Band interfaces must connect to VLAN aware equipment.**

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9. Configuring Wavence

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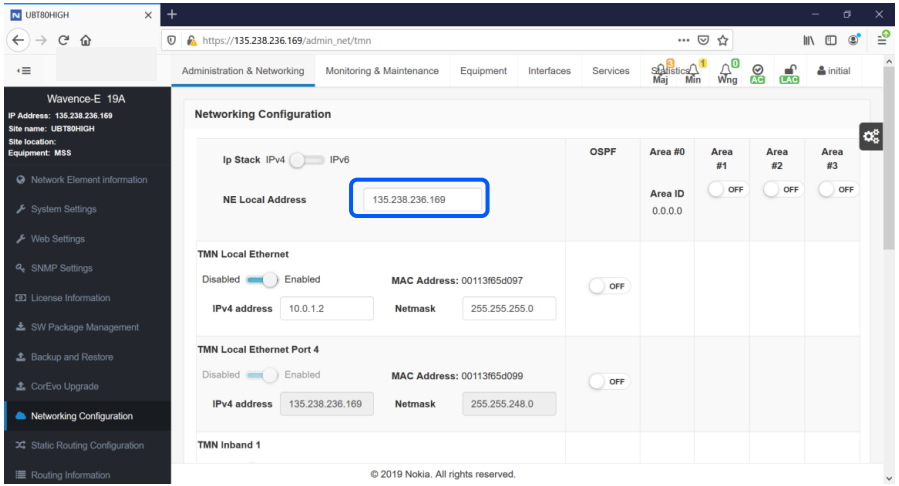
Wavence Networking

Wavence addressing – Setting the Local Address with WebCT



The Local Address is set in the “NE Local Address” field”.

Access the “NE Local Address” field by selecting:

1. Administration & Networking->Networking Configuration

Enter the Local Address and click Apply

1. The Local Address has an implied /32 netmask (255.255.255.255).
2. The Wavence will reboot after this address is changed.
3. After the reboot, you will need to reconnect using the new Local Address

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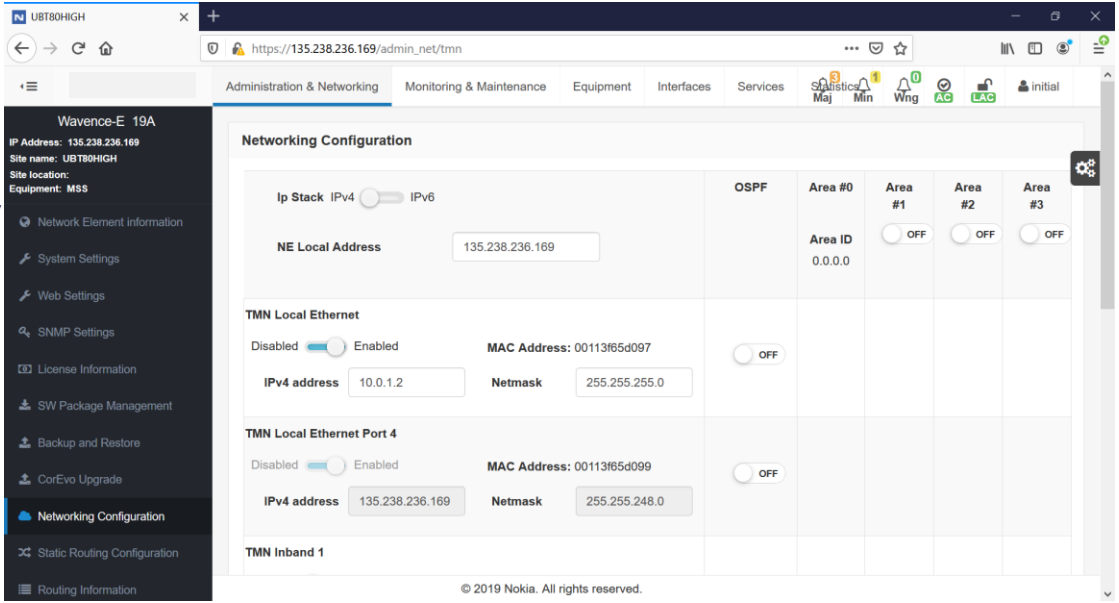
Wavence Networking

Wavence addressing – Configuring Port #4 for TMN mode with WebCT

Before Port #4 can be configured for TMN use any Services must be removed on this port.

1. Access the “TMN Local Ethernet Port 4” configuration from the Menu by selecting : Administration & Networking->Networking Configuration
2. Enable Port 4 for TMN (This will remove the port 4 from traffic ports)
3. Enter the Port 4 Address and netmask
4. Click Apply



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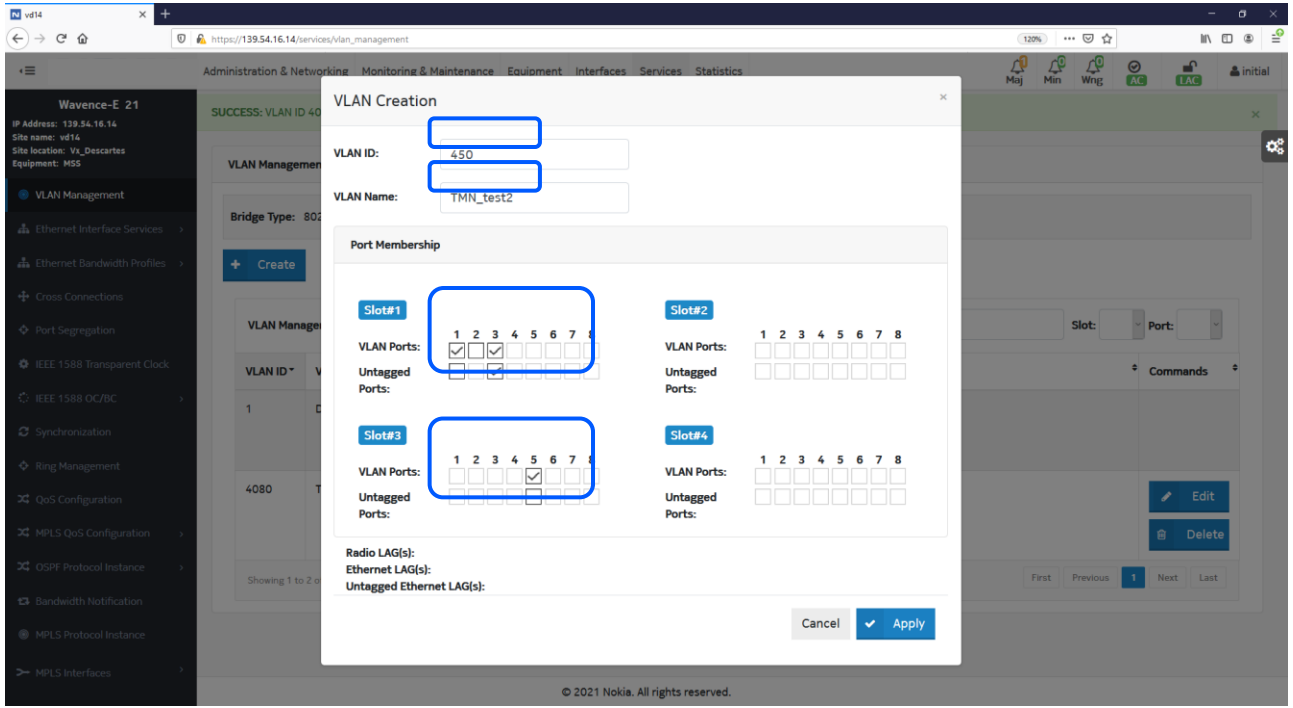
Wavence Networking

Wavence addressing – Configuring TMN IN band mode with WebCT

1. Create a VLAN :
   Services → Vlan Management → Create
2. Configure VLAN ID and VLAN Name
3. Select Associated Ports
4. Click Apply



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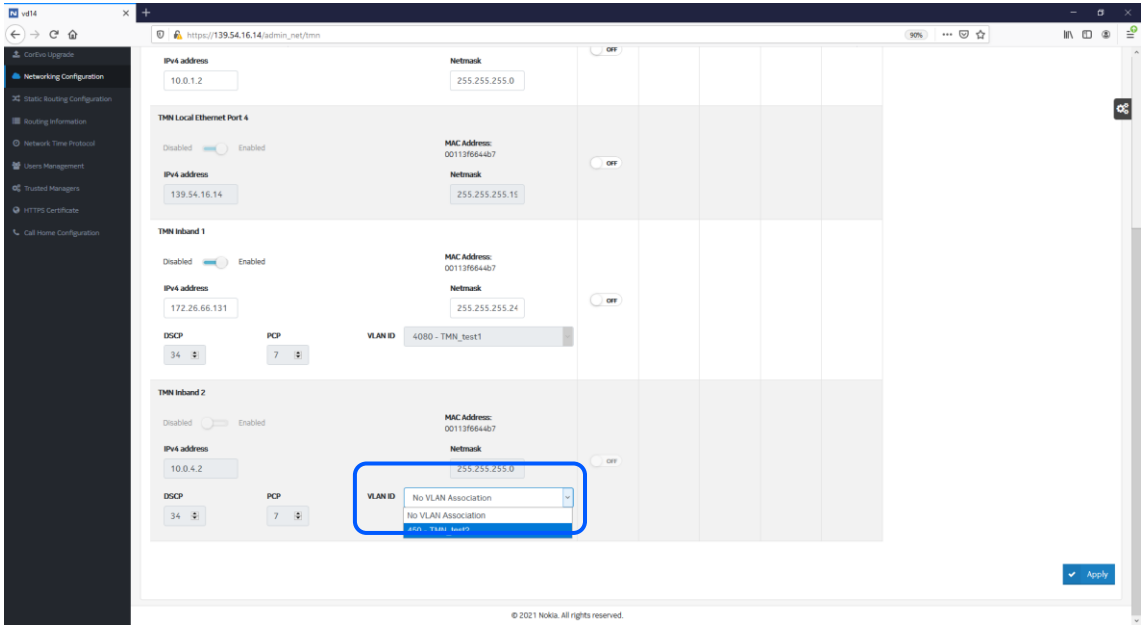
Wavence Networking

Wavence addressing – Configuring TMN IN band mode with WebCT

1. Access the “TMN inBand” configuration by selecting : Administration & Networking → Networking Configuration
2. Associate a VLAN with TMN Inband to configure



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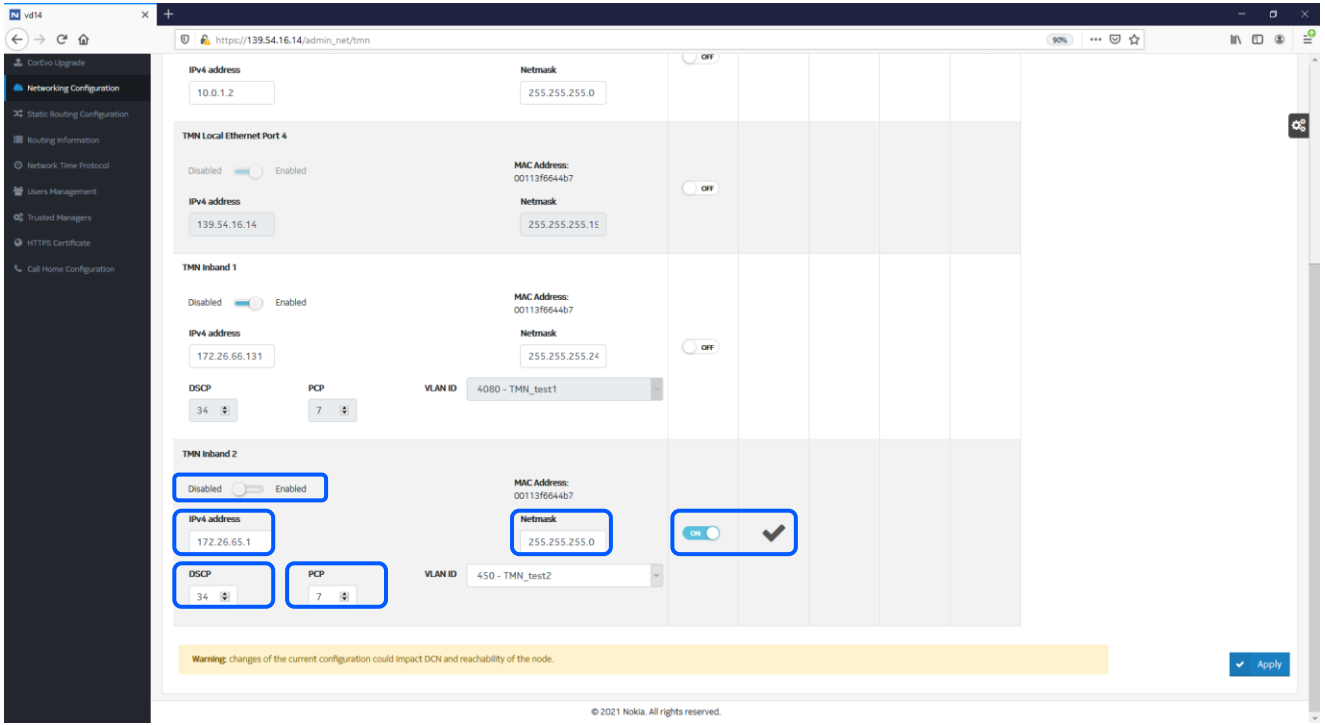
Wavence Networking

Wavence addressing – Configuring TMN IN band mode with WebCT

4. Modify DSCP or PCP if needed
5. Enter the TMNiB Address and netmask
4. Select OSPF routing if needed
5. Enable TMN inBand
6. Click Apply



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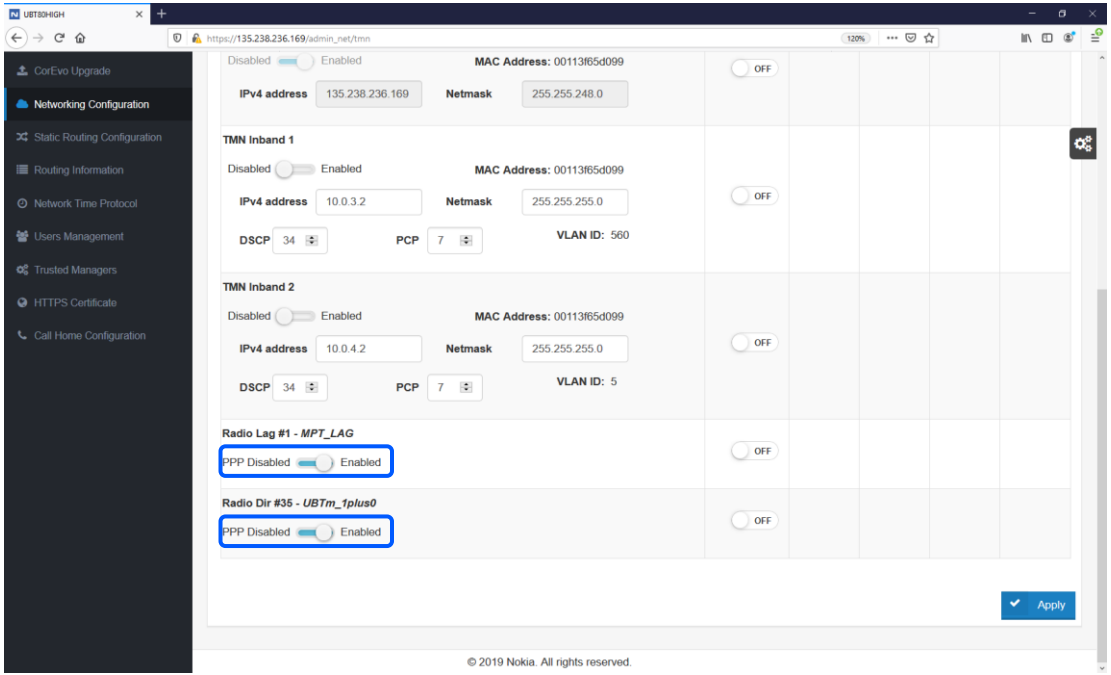
Wavence Networking

Wavence addressing – Configuring Radio RF PPPoE Links with WebCT

For RF PPPoE interfaces:

1. Access the “PPP” configuration from the Menu by selecting : Administration & Networking-> Networking Configuration
2. Enable PPP on desired Radio Port
3. Click on Apply



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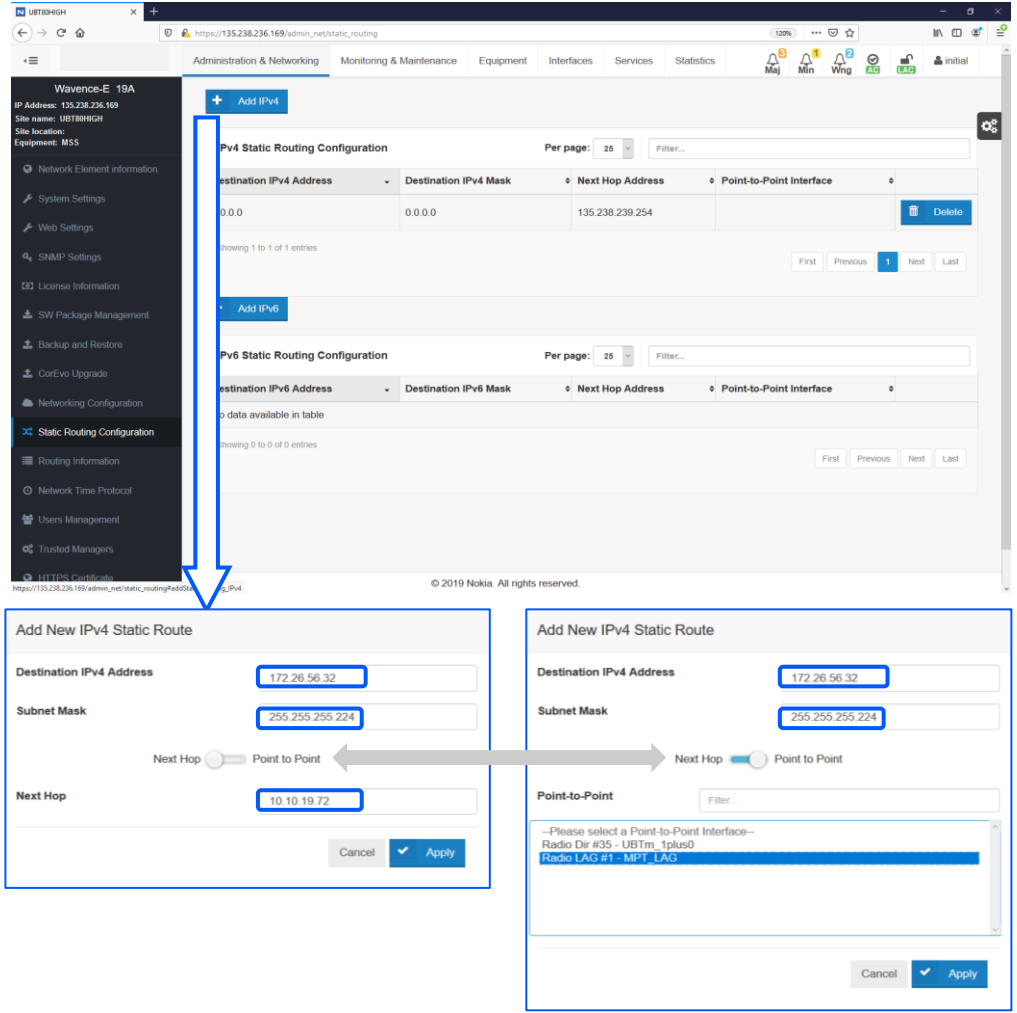
Wavence Networking

Wavence addressing – configuring static route with WebCT

Static Route Configuration:

1. Access the Menu by selecting Administration&Networking → Static Routing Configuration
2. To add a new static route click “Add IPv4” or “Add IPv6”.
3. Enter the destination subnet.
4. Configure the Interface Choice
    1. If the gateway is an IP address chose “Next Hop” and enter the address.
    2. If the gateway is a radio direction choose “Point to Point” and select the appropriate interface.
5. Gateways must always be on a directly attached network.
6. Click Apply.



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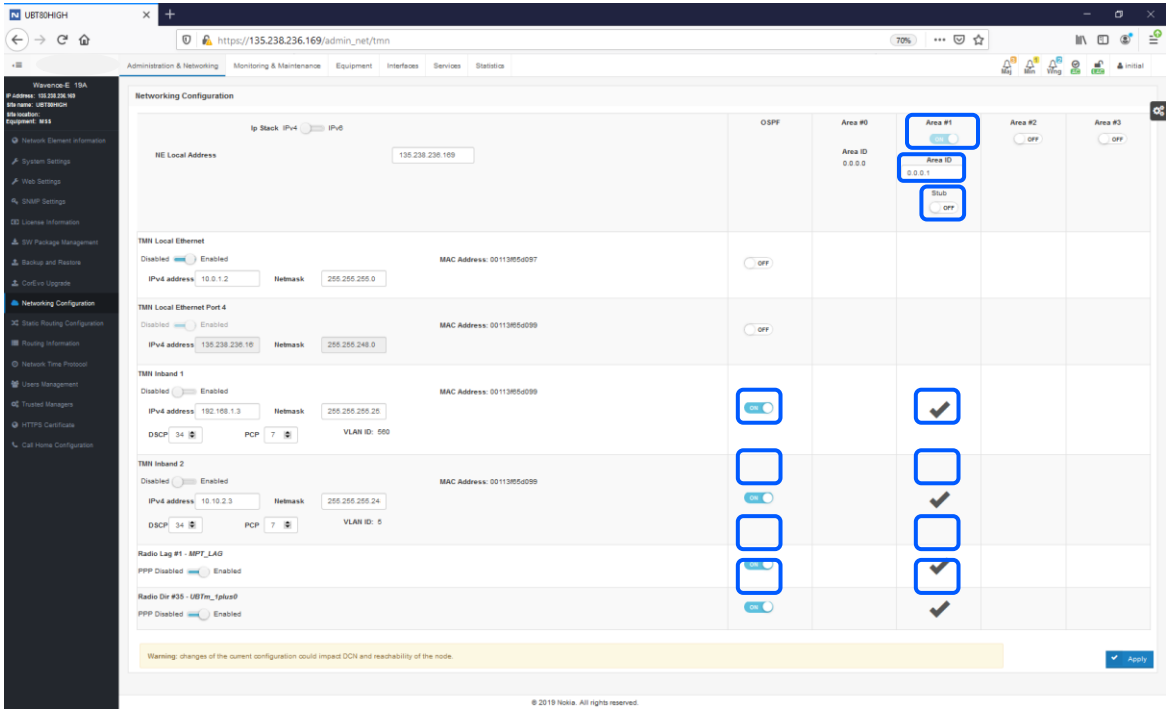
Wavence Networking

Wavence addressing – selecting on OSPF area for TMN interface with WebCT

1. Access the OPSF Configuration by selecting : Administration & Networking → Networking Configuration.
2. OSPF area 0.0.0.0 is created by default as Area#0.
3. Create a new area by selecting “ON” in a new area.
4. Fill the Area ID.
5. Select if Area is stub or no stub.
6. Select interfaces using OSPF routing by selecting “ON” in OSPF column.
   If “OFF” is selected Static routing will be use for this interface.
7. Select the right area for this interface.
8. Click Apply.



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10. TMN in UBT SA configurations

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UBT Standalone

TMN configurations (1/5)

1. The UBT SA supports TMN in-band with a provisionable VLAN tag. For local management the VLAN has to be dropped/inserted, on the port so that PC connection can be used without the need to have VLAN on the PC side.
2. The UBT SA supports also a dedicated local TMN port via the electrical RJ45 port. When using the local TMN port, VLANs are not supported.

3. There are 2 UBT SA user ports:
    1. one electrical port (RJ45 PoE) that is always enabled and is used only for TMN. The electrical port is non-configurable and is not monitored via alarms.
    2. one optical port that is always enabled and is used for traffic and TMN In Band. The optical port is configurable and monitored via alarms.
4. Note: on UBT-I the electrical port is only used for local management.

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UBT Standalone

TMN configurations (2/5)

1. Local Port Management: The UBT SA electrical port is reserved for out-of-band TMN management and therefore this port does not allow the processing of VLAN tagged frames. When VLAN tagged frames are sent through electrical interface, user data traffic loss may occur on the optical interface. To recover, it is recommended to perform a backup through WebCT of the running configuration, then perform a restore. The restore will trigger the Network Element restart and data traffic recovery. Please disconnect any traffic that has VLAN Tagged frames electrical interface after launching the restore process.

2. All TMN provisioning rules are applicable to local port management in order to access the remote node.

Note: on UBT-I, accessing the remote node through local port management is not possible.

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UBT Standalone

TMN configurations (3/5)

1. Flat IP TMN: UBT SA supports a Flat IP TMN architecture in which the TMN is not a routed network. In a Wavence link, both ends of the link are in the same IP subnet and each Wavence node is only assigned a single IP address for the control plane.

2. Unique IP address for TMN

3. TMN flows (in-band or out of band) are accessible through a single IP address: 192.168.100.1/24 which can be modified.

4. TMN In-band is associated to a specific VLAN (default 4080),which can be modified (range 2..4080).

5. For TMN Out-of-band, UBT inserts/drops TMN in-band VLAN, allowing this way PC connection without VLAN management.

6. Note: for UBT-I, the TMN-out-of-band IP address is fixed: 192.168.10.1

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UBT Standalone

TMN configurations (4/5)

TMN Provisioning: the unique working mode in UBT SA.

From IP stack and Networking perspective, UBT behaves as an IP host:

1. to provide host IP interface capabilities available in the OS.
2. to provision the IP address, Subnet Mask and default Gateway.
3. to provision the TMN In Band VLAN associated to the IP interface.

Note: No routing functionalities or routing protocols are required.

The TMN In-Band interface dedicated to TMN is used to connect UBT to LAN exchanging TMN information through a VLAN mixed with user Ethernet traffic.

It has its own IP address and subnet mask assigned via configuration by the management systems. The default configuration is:

- IP address: 192.168.100.1
- Network mask: 255.255.255.0
- TMN In-Band VLAN ID: 4080
- PCP: 7
- DSCP: 34
- Default Gateway: 0.0.0.0

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UBT Standalone

TMN configurations (5/5)

1. The IP Address must be configured according to the following rules
   1. Class A, Class B, Class C and classless addressing are allowed.
   2. Class D (Multicast) and Class E IPv4 addresses are not allowed.
   3. IP addresses from 127.0.0.0 to 127.255.255.255 are not allowed.
   4. IP Address 0.0.0.0 to 0.255.255.255.255 are not allowed.
   5. The IP sub-net mask is different from 255.255.255.254 (/31) and 255.255.255.255 (/32).
   6. IP Address must be checked with Subnet Mask in order to not allow IP address referring to the subnet itself (all 0 in the Host portion of the address) or referring to the subnet broadcast address (all 1 in the Host portion of the address).
2. **Network Element IP layer:** In TMN Provisioning management, NE local address and the IP of the TMN Inband interface have the same value. You can edit only the IP of TMN Inband, NE local address is read only. Flat IPv4 and IPv6 are supported.

NE interface configuration (IPv4/IPv6)

Name	Type	Default value
NE IP address	IPv4	192.168.100.1
NE subnet mask	IPv4 mask	255.255.255.0
NE IP address	IPv6	fc00:0:0:0:1::1

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UBT Standalone

TMN in band – IPv4

TMN Inband

The TMN Inband feature allows the extension of the DCN over User Ethernet interfaces used to carry user traffic. TMN Inband allows the user to manage a Wavence NE using general Ethernet port from a central location (NOC).

Inband TMN messages can reach the Wavence over the fiber port only. For Inband connectivity a VLAN is required.

The DSCP and PCP values allow a priority to be set for TMN management packets to the Data Communications Network (DCN). Queue mapping for TMN management packets is performed outside the Wavence. TMN packets inside the radio network are in a fixed high priority user traffic queue (Q7).

1. The DSCP value can range from 0 to 63. The default is 34.
2. The PCP value can range from 0 to 7. The default is 7.

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UBT Standalone

TMN in band – IPv6

TMN Inband – IPv6

Every time the TMN Inband IPv6 address is modified, the system performs a Duplicate Address Detection (DAD) check. When a duplicate address is detected, an alarm is raised, one for each TMN Inband networking interface. You need to change the IPv6 address to remove duplicates.

Note: Raising or clearing a DAD alarm is not successful when UBT is muted. DAD alarm is not raised during a cold restart or a power cycle.

IPv6 addresses FF02::1 (All nodes) and FF02::2 (All Routers) Multicast Group addresses must be joined by an enabled TMN Inband networking interface. NDP (Neighbor Discovery Protocol) is used to perform Neighbor Address Resolution to retrieve MAC address-IPv6 address association over the VLAN. IPv6 Stateless Address Autoconfiguration is not performed over TMN Inband networking interfaces.

Inband TMN Ethernet interface is automatically assigned 802.1p priority 0 on egress traffic.

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UBT Standalone

TMN in band – Provisioning

1 On the `Network Configuration` webpage, navigate to the *TMN Inband* field.

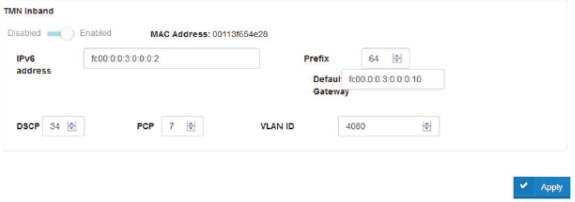
1.1 IPv4 stack

Figure 60: TMN Inband navigation - IPv4

Configure `IPv4 address`, `Netmask` and the `VLAN ID` of TMN in-band interface.

1.2 IPv6 stack

Figure 61: TMN Inband navigation - IPv6



1. Configure `IPv6 address`.
   Global Unicast address of TMN In-Band interface, default value `FC00:0:0:1::1`.
   Then NE local address aligns.
2. Configure `Prefix`.
   Prefix length of the IPv6 address. Default value: `64`. Range: `4` to `126`.

2 Choose the DSCP value between 0 and 63.

Default value: `34`.

3 Choose the PCP value from between 0 and 7.

Default value: `7`.

4 Choose the VLAN ID value.

Default value: `4080`.

5 Configure the Default Gateway.

No default gateway is assigned. Fill in an IPv4 or IPv6 address, depending on the configured IP stack.

The IP address value must belong to the same TMN subnet.

6 Click `Apply`.

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UBT Standalone

DHCP Server (1/3)

The support of DHCP server is applicable when UBT SA is working in IPv4.

In the UBT SA default configuration, the electrical Ethernet port is used as Management and the DHCP server on that port is enabled in standby mode and when the SA topology is confirmed.

The administrative setting of the DHCP server can be changed. The default IP addresses in the DHCP IP address pool are in the same subnet of the default TMN in-band IP address (192.168.100.0/24). There are 4 default IP addresses in the pool: 101, 102, 103 and 104. They are the same for all UBT SA.

Note: the UBT-I DHCP server only concerns Out-of-band. For the Out-of-band the Ips provided will be only that of the current debug port. DHCP will be always active for the local management port.

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UBT Standalone

DHCP Server (2/3)

The default DHCP configuration is as follows:

1. When the TMN OoB is enabled, the DHCP Server is enabled by default.
2. When the TMN OoB is disabled and enabled again, the DHCP server is disabled by default.
3. In case of SW migration from previous release where DHCP configuration is not supported and with the TMN in-band IP address different to the default one of release W21A, where DHCP configuration is allowed, the DHCP server is disabled by default since the configuration of TMN In-Band IP (coming from previous configuration) and the configuration of DHCP Pool (set as default as we are coming from a SWP Activation of W21A) are not satisfying the IP address consistency checks.

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UBT Standalone

DHCP Server (3/3)

For UBT SA CA:

1. When DHCP client is connected to the Primary UBT, the DHCP discover broadcast message is forwarded to the Secondary UBT over the coupling link. Therefore, the client receives DHCP offers from the 4 UBTs

2. When DHCP client is connected to the Secondary UBT, only the Secondary UBT responses with DHCP offer message to the client. The secondary does not forward the DHCP discovery broadcast frame to the coupling. The client receives a single DHCP.

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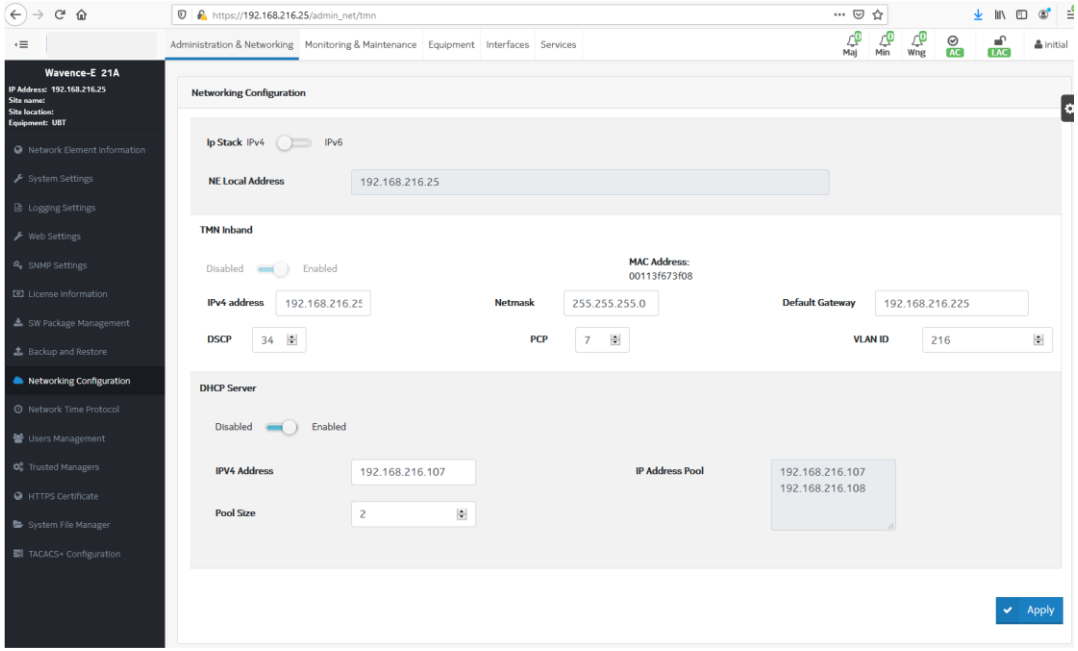
UBT Standalone

DHCP Server configuration

From the Administration & Networking domain, select Networking Configuration.
Disable or enable the DHCP server and configure the IPv4 Address field.

Note: the DHCP Server part on WebCT is not visible on UBT-I



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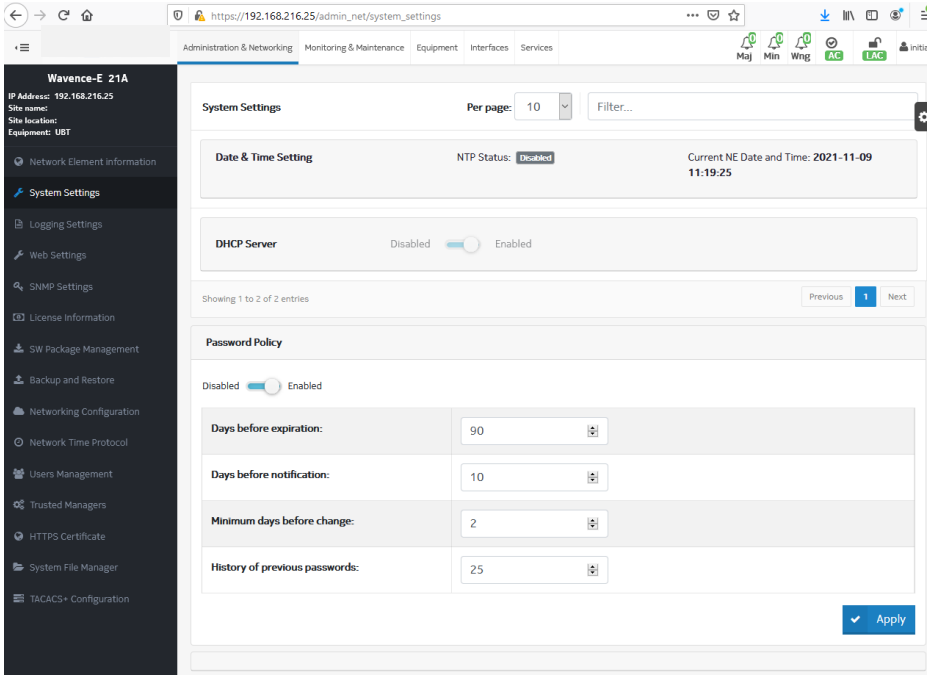
UBT Standalone

DHCP Server administrative status

From the Administration & Networking domain, select System Settings.

Note: On UBT-I, the DHCP server is always enabled and not configurable.



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11. Default and Reserved addresses

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Wavence TMN Networking

Default address

MSS-4/8 Shelf, MSS-E/HE/XE

IPv4

- 10.0.1.2 NE Local Address
- 10.0.1.2/24 TMN Ethernet port
- 10.0.2.2/24 Port #4 in TMN Mode (MSS-4/8 only)
- 10.0.3.2/24 TMN In-Band #1
- 10.0.4.2/24 TMN In-Band #2

IPv6

- FEC0:0:0:1::1/128 Local Address
- FEC0:0:0:1::1/64 TMN Ethernet port
- FEC0:0:0:2::1/64 Port #4 in TMN Mode (MSS-4/8 only)
- FEC0:0:0:3::1/64 TMN In-Band #1
- FEC0:0:0:4::1/64 TMN In-Band #2

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Wavence TMN Networking

Internal address

The following address ranges are used internally by Wavence. Addresses from the following ranges should not be assigned as a Local Address or used for an interface address.

MSS:

- 127.0.0.0/8 Loopback
- 100.1.0.0/24 Internal Core Main to Core Spare communications
  - 100.1.0.1 Core Main
  - 100.1.0.2 Core Spare
- 100.2.0.0/24 Internal MSS to Radio communications
  - 100.2.0.1 Core
  - 100.2.0.x one for each Radio, depending on slot/port

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Wavence TMN Networking

MTUs

TMN Interfaces do not support jumbo frames.

The MTUs:

- TMN PPP RF links
  - 1492 Byte Layer 2 MTU
- TMN Broadcast Ethernet interfaces
  - 1500 Byte Layer 3 MTU
  - 1518 Byte Layer 2 MTU
  - 1522 Byte Layer 2 MTU for TMN In-Band with 802.1q VLAN tag.

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12. Known issues

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Known Issues

There is a potential for intermittent route failure when the Wavence Local Address matches a port address. This issue affects all releases from MPR R6.0 through Wavence 18. The issue has been fixed in Wavence 18A

Conditions that can trigger the problem:

1. The Wavence Local Address matches the TMN port address of one of the local Ethernet interfaces. - This addressing method is allowed by the Wavence TMN networking rules.
2. The Wavence Local Address is contacted from a remote IP address using a TCP based protocol (such as HTTP, HTTPS FTP, SFTP, etc). A known trigger is using a Web Browser to log into a remote Wavence system. SNMP (UDP) does not trigger the problem but may be affected.
3. Replies to the TCP connection would be returned through a gateway in the subnet where the Wavence Port address matches the Local Address.

When all of the above conditions are met a temporary local routing loop in the Wavence NE may be triggered (XHAULEXE-11477). The temporary routing loop affects all TMN communication with that NE. TMN communications that are routed through the Wavence to and from other remote devices are NOT affected. Once the TCP flow stops, routing returns to normal. Refer to the examples on the next slides.

WORKAROUND: Change the Wavence addressing so the Local Address does not match any of the local TMN ethernet interface ports.

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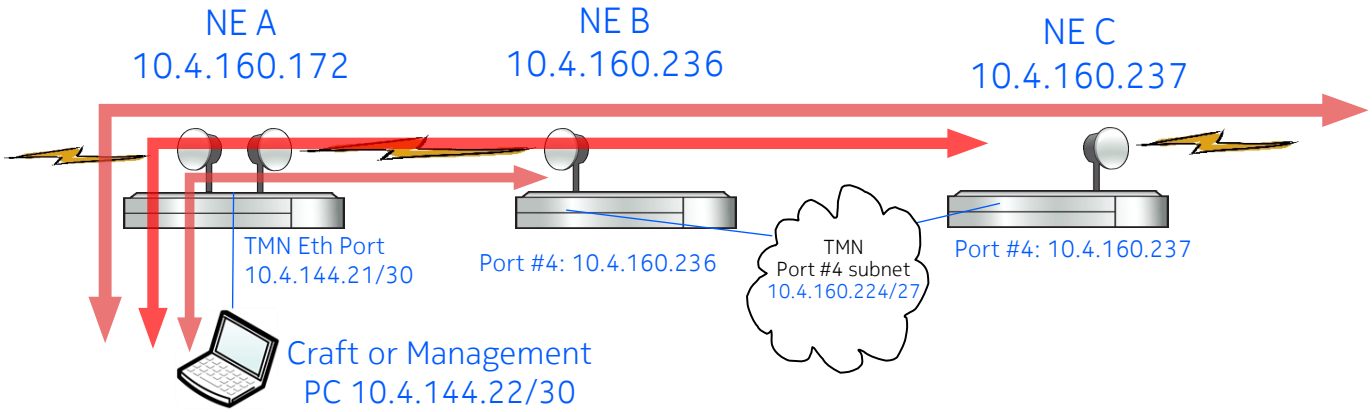
Known issue: Scenario 1



1. In this topology Wavence Local Addresses at NE B and NE C match the TMN Port #4 interface addresses.
2. When using a web browser (HTTP) connection from the Laptop:
    1. to A and B work as expected.
    2. to sites beyond C work as expected
    3. Communication to C will be very slow or fail completely. TCP replies from the Local Address/TMN Port #4 at C use the TMN Port #4 gateway of B triggering the temporary routing loop issue.
3. Workaround: Change the Wavence addressing at B and C so the Local Addresses are outside the TMN Port #4 subnet.

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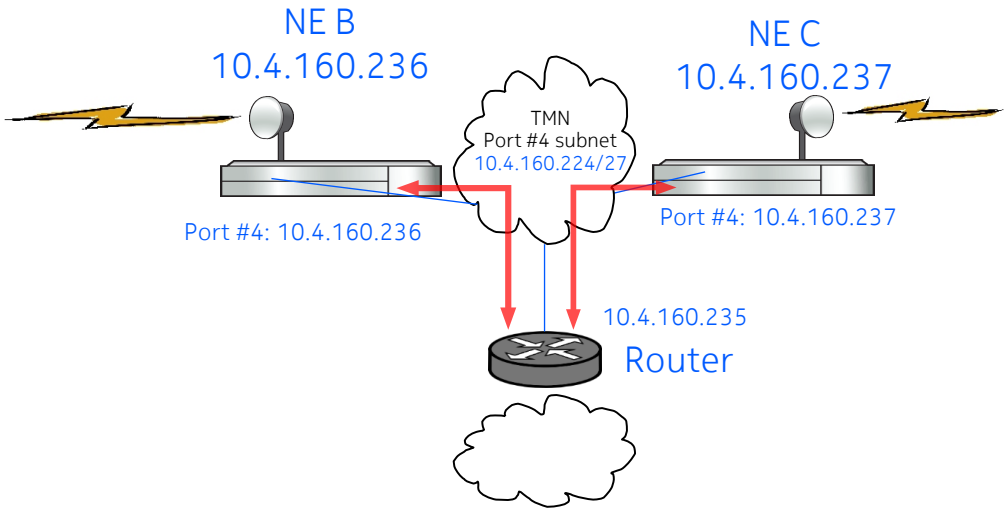
Known issue: Scenario 2



1. In this topology Wavence Local Addresses at NE B and NE C match the TMN Port #4 interface addresses and the router has an interface in the same subnet.
    1. TCP Communication with B or C will be very slow or fail completely. TCP replies from the Local Address/TMN Port #4 at B or C using the router interface as the gateway can trigger the temporary routing loop issue. The temporary routing loop can disrupt other communications. OSPF can be affected causing route distribution problems.
2. Workaround: Change the Wavence addressing at B and C so the Local Addresses are outside the TMN Port #4 subnet.

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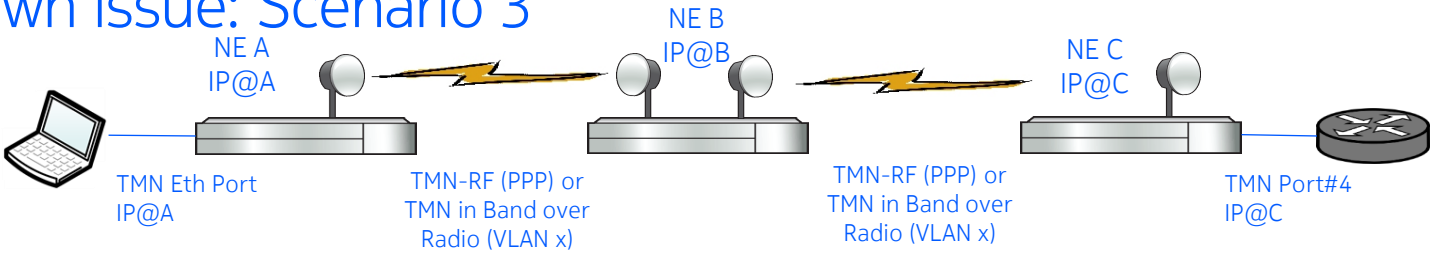
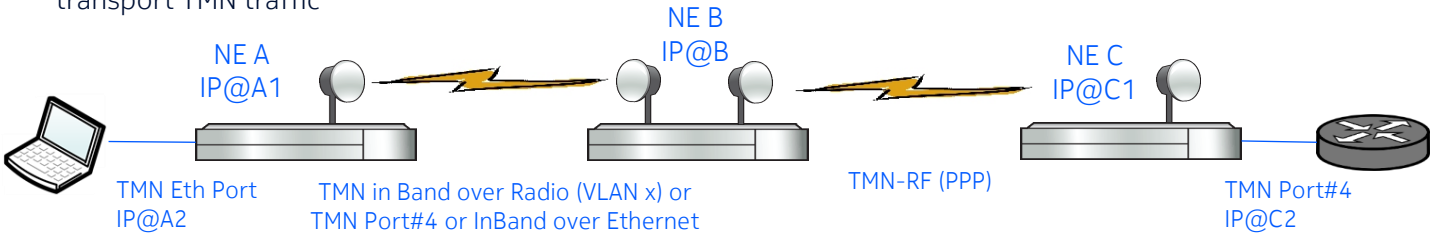
Known issue: Scenario 3



1. In IPv6 networks, it is allowed to assign the same IP address to Local IP and other external TMN interfaces (e.g., TMN Ethernet Port). This is applicable to:
    1. Networks using TMN-RF (based on PPP) on radio interfaces and OSPFv3 to signal IP prefix reachability
    2. Network design according to a Flat IP approach, having all the nodes in the same subnet and using TMN in band over Radio to transport TMN traffic



1. Whenever the IPv6 network is based on a hybrid concatenation of TMN transport technologies, it's recommended to assign different IP addresses to the Local IP and other TMN interfaces. The user might experience reachability issues (e.g., the laptop user in the example not being able to reach NE B from NE A during I&C) due to some limitation in the OSPFv3 protocol to advertise those overlapping routes

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BASIC NETWORK TRAINING

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A. Planning and Addressing a Network

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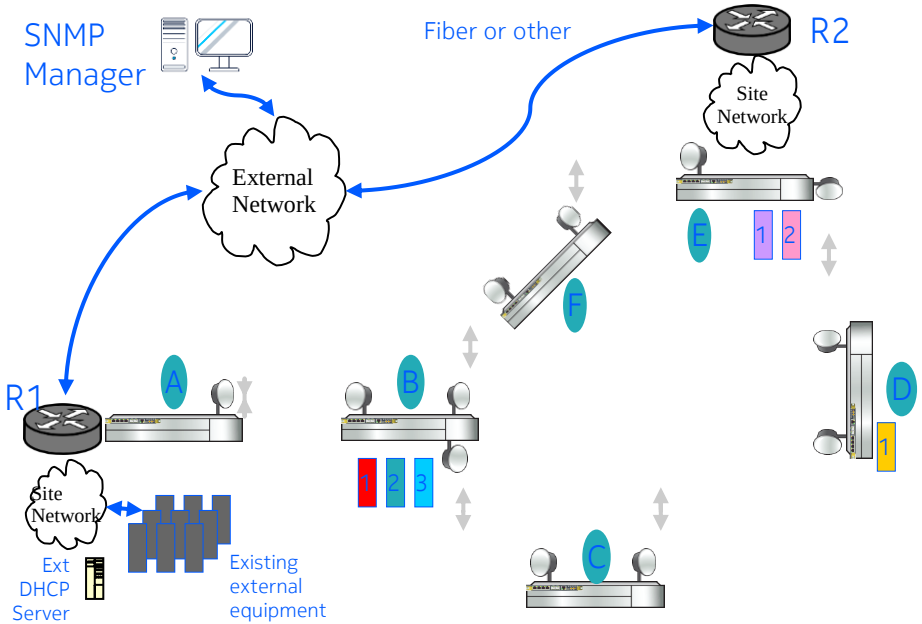
Wavence TMN Networking

TMN Network Planning



1. Consider the above network. For TMN purposes, it is a combination of Linear and Ring topologies.
2. External equipment [1 2 3]]to be managed is located at sites B, D, and E.
3. How could this network be addressed?

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Wavence TMN Networking

TMN Network Planning (1/4)

Definitions for the Example:

1. The type of addresses needed will depend on whether we use IPv4 or IPv6 and the type of connectivity required with external networks.
2. If the management network is private we can use address space intended for private networks.
    1. For IPv4 addresses for private networks are defined in RFC1918:
        - 10.0.0.0 /8
        - 172.16.0.0 /12
        - 192.168.0.0 /24
    2. For IPv6 they are defined in RFC4193
        - IPv6 addresses are typically assigned from the fd00:: /8 sub-range (L-bit set to 1). A suitable Locally Assigned Global ID will be required. Refer to the RFC for more information.

For this example we will use IPv4 with Private Addressing

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Wavence TMN Networking

TMN Network Planning (2/4)

1. Site A is an existing site where other equipment is already installed. A Wavence device will be added to the site. The existing subnet is 192.168.19.0/27. Router R1 gateway is at 192.168.19.1. The address available for Wavence is 192.168.19.23.
2. Site B is a junction. There are three external devices to be managed via the TMN Network.
3. Site C is a repeater with no external equipment.
4. Site D is a repeater with one external device.
5. Site E is a repeater with two external devices. A new router (R2) connected via an external link will be added to provide an alternate pathway for TMN Traffic. The Wavence DHCP server will be used to provide DHCP services for direct craft connections at this site.
6. Site F is a repeater with no external equipment.

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Wavence TMN Networking

TMN Network Planning (3/4)

1. OSPF will be enabled within the network
    1. The recommended configuration is to enable OSPF within the Wavence network whenever possible.
        1. Correctly configuring Static routing internally within the Wavence network can be very complex for anything other than trivial linear networks and is not recommended.
2. The Wavence network will be an autonomous OSPF network. It will use Static routing at the borders
    1. The use of OSPF or Static routing between the Wavence network border and external networks is a network design choice. When OSPF monitors the status of a link carrying TMN Traffic and the link fails it may be possible for OSPF to reroute the TMN traffic to use an alternate gateway provided one is available.

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Wavence TMN Networking

TMN Network Planning (4/4)

1. At site A where an external network is available, Wavence will be a member of the external network, using the Port #4 TMN interface. Wavence will be configured to use router R1 as the Default Gateway for reaching all other external networks.
2. At site E, a new local network will be defined using router R2. This network will provide an alternate external route for TMN traffic. The Wavence at site E will be come a member of this network and will use Router R2 as the Default Gateway to reach the external networks
3. The Wavence Local Address will be set to match the TMN Ethernet Port address at all sites.
4. DHCP will be used for Craft computers connected to the local TMN Ethernet port.
    1. Using the Wavence internal DHCP server whenever possible is recommended. The internal DHCP server will correctly configure external Craft computers to communicate with the local Wavence and the greater TMN Network. This eliminates the need for users to know how to manually configure a laptop at each site.

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Wavence TMN Networking

TMN Network Planning - New Address requirement

Site	TMN Ethernet Network Interface	Port #4 TMN Network Interface	Comments
Site A	4 address subnet (/30)	0 ***new*** addresses (Wavence assigned an available address from the existing external network)	Port #4 TMN interface will be connected with the local site network. The separate TMN Ethernet port will remain active with DHCP direct Local access.
Site B	4 address subnet (/30)	8 address subnet (/29)	Connect external equipment to the Port #4 TMN interface.
Site C	4 address subnet (/30)	(disabled)	No external equipment. Local direct craft connections only.
Site D	8 address subnet (/29)	(disabled)	A very small local network. One external device connected to the TMN Eth port. There is no addressing advantage in splitting the subnet between TMN Eth and Port #4 TMN
Site E	16 address subnet (/28)	(disabled)	Extra addresses reserved to allow adding equipment as the network expands.
Site F	4 address subnet (/30)	(disabled)	No external equipment. Direct craft connections only.
Total	**40 addresses**	**8 addresses**	

1. The TMN Ethernet port with DHCP enabled will be active at all sites.
2. The Port #4 TMN interface is only enabled at two sites: A and B
    1. The Addressing for the Port #4 interface at Site A is not shown here. This interface is connected to and uses a spare address from the existing external site network. This was stated in the Definitions near the beginning of this example.
3. At sites where external equipment is present it is connected to the Wavence using an external switch.

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Wavence TMN Networking

TMN Network Planning

1. With the network plan shown in the previous table, this network requires the space of at least 48 addresses.

2. Of the 48:
   1. 6 addresses are assigned as Wavence Local Address/TMN Ethernet addresses
   2. 1 address is assigned as a TMN Port #4 address
   3. 7 addresses are assigned to external equipment (including router R2)
   4. 14 addresses are reserved for use by the internal Wavence DHCP servers
   5. 6 Spare addresses

3. Netmasks require subnets to start and end on specific boundaries. For example a subnet of 8 addresses must start at an address that is evenly divisible by 8.

4. Before we request address space from our Network Administrator, we first need to verify what size address block will hold all our subnets. This is shown on the next slide.

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Wavence TMN Networking

TMN Network Planning

Wavence A TMN Eth Subnet	
Netmask: 255.255.255.252	
xxx.xxx.xxx.0	Network (/30)
xxx.xxx.xxx.1	**A TMN Eth Port**
xxx.xxx.xxx.2	DHCP Assigned
xxx.xxx.xxx.3	Broadcast
Wavence B TMN Eth Subnet	
Netmask: 255.255.255.252	
xxx.xxx.xxx.4	Network (/30)
xxx.xxx.xxx.5	**B TMN Eth Port**
xxx.xxx.xxx.6	DHCP Assigned
xxx.xxx.xxx.7	Broadcast
Wavence B Port #4 TMN Subnet	
Netmask: 255.255.255.248	
xxx.xxx.xxx.8	Network (/29)
xxx.xxx.xxx.9	**B Port #4**
xxx.xxx.xxx.10	B Ext Eqpt 1
xxx.xxx.xxx.11	B Ext Eqpt 2
xxx.xxx.xxx.12	B Ext Eqpt 3
xxx.xxx.xxx.13	Spare
xxx.xxx.xxx.14	Spare
xxx.xxx.xxx.15	Broadcast

Wavence C TMN Eth Subnet	
Netmask: 255.255.255.252	
xxx.xxx.xxx.16	Network (/30)
xxx.xxx.xxx.17	**C TMN Eth Port**
xxx.xxx.xxx.18	DHCP Assigned
xxx.xxx.xxx.19	Broadcast
Wavence F TMN Eth Subnet	
Netmask: 255.255.255.252	
xxx.xxx.xxx.20	Network (/30)
xxx.xxx.xxx.21	**F TMN Eth Port**
xxx.xxx.xxx.22	DHCP Assigned
xxx.xxx.xxx.23	Broadcast
Wavence D TMN Eth Subnet	
Netmask: 255.255.255.248	
xxx.xxx.xxx.24	Network (/29)
xxx.xxx.xxx.25	DHCP Assigned
xxx.xxx.xxx.26	DHCP Assigned
xxx.xxx.xxx.27	DHCP Assigned
xxx.xxx.xxx.28	**D TMN Eth Port**
xxx.xxx.xxx.29	D Ext Eqpt 1
xxx.xxx.xxx.30	Spare
xxx.xxx.xxx.31	Broadcast

Wavence E TMN Eth Subnet	
Netmask: 255.255.255.240	
xxx.xxx.xxx.32	Network (/28)
xxx.xxx.xxx.33	DHCP Assigned
xxx.xxx.xxx.34	DHCP Assigned
xxx.xxx.xxx.35	DHCP Assigned
xxx.xxx.xxx.36	DHCP Assigned
xxx.xxx.xxx.37	DHCP Assigned
xxx.xxx.xxx.38	DHCP Assigned
xxx.xxx.xxx.39	DHCP Assigned
xxx.xxx.xxx.40	**E TMN Eth Port**
xxx.xxx.xxx.41	E Ext Eqpt 1
xxx.xxx.xxx.42	E Ext Eqpt 2
xxx.xxx.xxx.43	Spare
xxx.xxx.xxx.44	Spare
xxx.xxx.xxx.45	Spare
xxx.xxx.xxx.46	Router R2
xxx.xxx.xxx.47	Broadcast

This shows how the required subnets could map into an arbitrary block of 48 contiguous addresses.
Other arrangements are possible. Notice the addresses for Wavence F are between the addresses of Wavence C and Wavence D.
With OSPF enabled there will be no routing complications resulting from this choice.
Also notice TMN Eth Port addresses for D and E are in the middle of the subnet.
This is to limit addresses reserved by DHCP.
Refer to Appendix C for more information.

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Wavence TMN Networking

TMN Network Planning

1. The previous slide shows how this network design will fit within the total space of 48 addresses. This is the minimum sized block number of addresses we must request to deploy this network.

2. Depending on the availability of IP address space:
    1. we could be assigned a single /26 block with 64 addresses.
    2. If the availability of new address space is limited, we might be assigned exactly 48 addresses broken down as:
        1. a /27 block (32 addresses) plus
        2. a /28 (16 addresses)
    3. If assigned separately like this, the address blocks might not be contiguous. With OSPF enabled no special configuration for routing is required.

3. For this example, we'll assume we've been assigned 172.28.137.64 /26
    1. The block contains 64 contiguous addresses ranging from 172.28.137.64 to 172.28.137.127
    2. We'll further divide this block of addresses into our own subnets following the plan.

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Wavence TMN Networking

TMN Network Planning

Wavence A TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.64	Network (/30)
172.28.137.65	**A TMN Eth Port (L)**
172.28.137.66	DHCP Assigned
172.28.137.67	Broadcast

Wavence B TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.68	Network (/30)
172.28.137.69	**B TMN Eth Port (L)**
172.28.137.70	DHCP Assigned
172.28.137.71	Broadcast

Wavence B Port #4 TMN Subnet	
Netmask: 255.255.255.248	
172.28.137.72	Network (/29)
172.28.137.73	**B Port #4**
172.28.137.74	B Ext Eqpt 1
172.28.137.75	B Ext Eqpt 2
172.28.137.76	B Ext Eqpt 3
172.28.137.77	Spare
172.28.137.78	Spare
172.28.137.79	Broadcast

Wavence C TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.80	Network (/30)
172.28.137.81	**C TMN Eth Port (L)**
172.28.137.82	DHCP Assigned
172.28.137.83	Broadcast

Wavence F TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.84	Network (/30)
172.28.137.85	**F TMN Eth Port (L)**
172.28.137.86	DHCP Assigned
172.28.137.87	Broadcast

Wavence D TMN Ethernet Subnet	
Netmask: 255.255.255.248	
172.28.137.88	Network (/29)
172.28.137.89	DHCP Assigned
172.28.137.90	DHCP Assigned
172.28.137.91	DHCP Assigned
172.28.137.92	**D TMN Eth Port (L)**
172.28.137.93	D Ext Eqpt 1
172.28.137.94	Spare
172.28.137.95	Broadcast

Wavence E TMN Ethernet Subnet	
Netmask: 255.255.255.240	
172.28.137.96	Network (/28)
172.28.137.97	DHCP Assigned
172.28.137.98	DHCP Assigned
172.28.137.99	DHCP Assigned
172.28.137.100	DHCP Assigned
172.28.137.101	DHCP Assigned
172.28.137.102	DHCP Assigned
172.28.137.103	DHCP Assigned
172.28.137.104	**E TMN Eth Port (L)**
172.28.137.105	E Ext Eqpt 1
172.28.137.106	E Ext Eqpt 2
172.28.137.107	Spare
172.28.137.108	Spare
172.28.137.109	Spare
172.28.137.110	Router R2
172.28.137.111	Broadcast

Wavence A TMN Port #4 Subnet	
Netmask: 255.255.255.224	
192.168.19.1	Router R1
192.168.19.23	Port #4

Merging our assigned addresses into the tables shows how to address our equipment.

Wavence Local addresses will be set to match the TMN Ethernet Port as specified in the Plan and are labeled (L) for easy identification.

Address assignments from the existing external network at Site A.

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Wavence TMN Networking

TMN Network Planning

172.28.137.112	Unused
172.28.137.113	Unused
172.28.137.114	Unused
172.28.137.115	Unused
172.28.137.116	Unused
172.28.137.117	Unused
172.28.137.118	Unused
172.28.137.119	Unused
172.28.137.120	Unused
172.28.137.121	Unused
172.28.137.122	Unused
172.28.137.123	Unused
172.28.137.124	Unused
172.28.137.125	Unused
172.28.137.126	Unused
172.28.137.127	Unused

16 addresses out of the 64 allocated addresses remain unused.

These will be reserved for future expansion of the network.

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Wavence TMN Networking

TMN Network Planning

According to the plan:

1. The Wavence at Site A will be configured as an ASBR with a static route to pointing to R1 as the default gateway. This is required so that TMN traffic destined to leave the Wavence TMN Network will have a way out

2. Conversely, External router R1 must be configured to use Wavence A as a gateway to reach network 172.28.137.64 /26

3. The Wavence at site E will be configured as an ASBR with a static route to use R2 as it's default gateway. This defines an alternate gateway for traffic leaving the Wavence TMN network.

4. External router R2 must also be configured to use Wavence E as a gateway to reach network 172.28.137.64 /26
   1. This route provides an alternate way into the Wavence TMN Network.
   2. Routing in the external DCN between R1 and R2 is beyond the scope of this presentation

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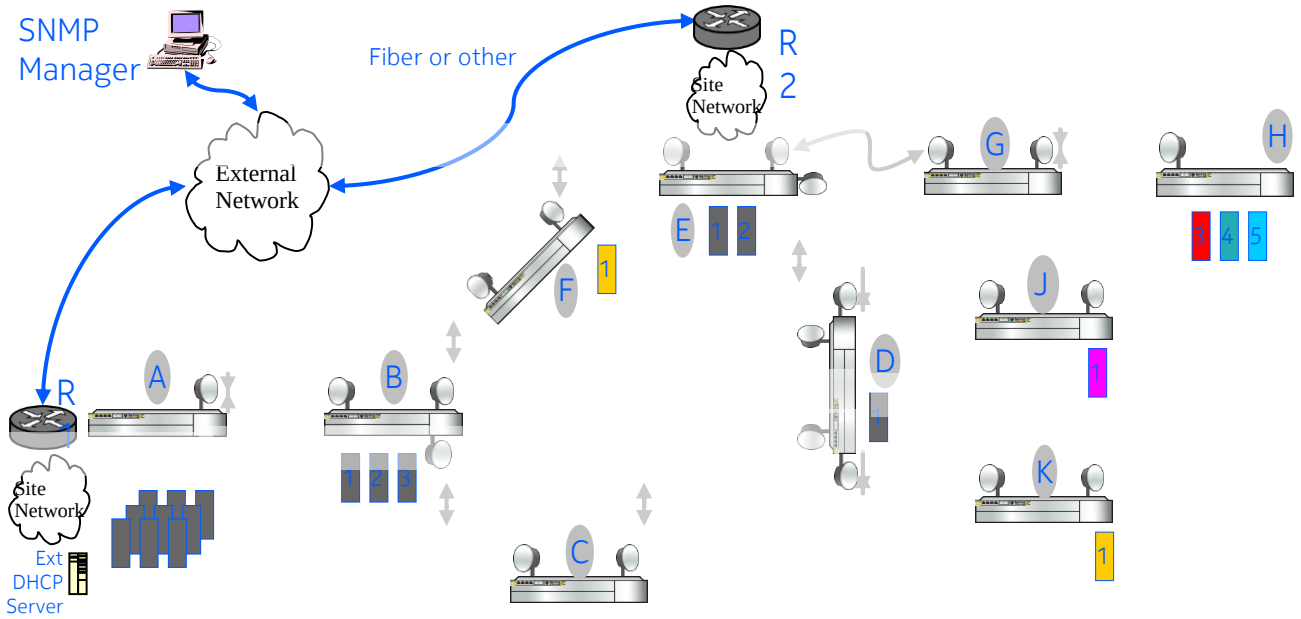
Wavence TMN Networking

TMN Network Planning – Growing the network



1. What if the network grows and we need to expand by adding Sites G, H, J, and K?
2. What if Site F, where we originally had no plans for expansion suddenly needs a new external device?
3. What if the old expansion plans change, and the additional external equipment planned for site E will now be deployed at site H, leaving site E with an excess of unused addresses?
4. We can use some of the sixteen addresses left over from the original /26, but we don't have enough addresses for all new the equipment. We'll have to request additional address space.

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Wavence TMN Networking

TMN Network Planning – Growing the network

Here's the new plan:

Site	TMN Ethernet Network	Port #4 TMN Network	Comments
Site E	Original 16 address subnet now a **4 addresses subnet (/30)**	8 address subnet (/29) (local reallocation)	Extra address space no longer needed. We split the 16 addresses subnet, allocating 8 addresses to **Port #4 TMN** and 4 addresses to **TMN Ethernet**. The four *recovered* addresses are moved to site F.
Site F	(Keep existing subnet)	4 address subnet (/30) (reallocated addresses from Site E)	Add subnet to **Port #4 TMN** interface for new equipment, using the address space recovered from Site E.
Site G	4 address subnet (/30)	(disabled)	No external equipment. Direct craft connections only.
Site H	4 address subnet (/30)	8 address subnet (/29)	External eqpt connected to **Port #4 TMN** interface.
Site J	8 address subnet (/29)	(disabled)	
Site K	8 address subnet (/29)	(disabled)	
Total	**24 new addresses**	**8 new addresses**	

1. Expansion of the network will require the space of 32 addresses, 16 of which were left over from our original allocation, and 16 new addresses.
2. We request a new block of 16 addresses from our Network Administrator:
3. Our new address block assigned is: 172.30.10.0 /28

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Wavence TMN Networking

TMN Network Planning – Growing the network

Wavence A TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.64	Network (/30)
172.28.137.65	**A TMN Eth Port (L)**
172.28.137.66	DHCP Assigned
172.28.137.67	Broadcast

Wavence B TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.68	Network (/30)
172.28.137.69	**B TMN Eth Port (L)**
172.28.137.70	DHCP Assigned
172.28.137.71	Broadcast

Wavence B Port #4 TMN Subnet	
Netmask: 255.255.255.248	
172.28.137.72	Network (/29)
172.28.137.73	**B Port #4**
172.28.137.74	B Ext Eqpt 1
172.28.137.75	B Ext Eqpt 2
172.28.137.76	B Ext Eqpt 3
172.28.137.77	Spare
172.28.137.78	Spare
172.28.137.79	Broadcast

Wavence C TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.80	Network (/30)
172.28.137.81	**C TMN Eth Port (L)**
172.28.137.82	DHCP Assigned
172.28.137.83	Broadcast

Wavence D TMN Ethernet Subnet	
Netmask: 255.255.255.248	
172.28.137.88	Network (/29)
172.28.137.89	DHCP Assigned
172.28.137.90	DHCP Assigned
172.28.137.91	DHCP Assigned
172.28.137.92	**D TMN Eth Port (L)**
172.28.137.93	D Ext Eqpt 1
172.28.137.94	Spare
172.28.137.95	Broadcast

Wavence F TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.84	Network (/30)
172.28.137.85	**F TMN Eth Port (L)**
172.28.137.86	DHCP Assigned
172.28.137.87	Broadcast

With the new expanded network plan, original subnets at site A, B,C, D, and the TMN Ethernet subnet at site F remain unchanged. The addresses are repeated here:

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Wavence TMN Networking

TMN Network Planning

This is how the old subnet E can be split to recover some unused address space. Half the address space with the external equipment moves to the Site E Port #4 TMN subnet with a new netmask. Four addresses from the original subnet remain assigned to the Wavence E TMN Eth subnet with a new netmask. The remaining four addresses are relocated to Site F as a new Port #4 subnet. New or changed addressing parameters are highlighted.

OLD

Wavence E TMN Ethernet Subnet	
Netmask: 255.255.255.240	
172.28.137.96	Network (/28)
172.28.137.97	DHCP Assigned
172.28.137.98	DHCP Assigned
172.28.137.99	DHCP Assigned
172.28.137.100	DHCP Assigned
172.28.137.101	DHCP Assigned
172.28.137.102	DHCP Assigned
172.28.137.103	DHCP Assigned
172.28.137.104	**E TMN Eth Port (L)**
172.28.137.105	E Ext Eqpt 1
172.28.137.106	E Ext Eqpt 2
172.28.137.107	Spare
172.28.137.108	Spare
172.28.137.109	Spare
172.28.137.110	Router R2
172.28.137.111	Broadcast

NEW

Wavence E TMN Ethernet Subnet	
Netmask: 255.255.255.248	
172.28.137.96	Network (/30)
172.28.137.97	**Wavence E TMN Eth (L)**
172.28.137.98	DHCP Assigned
172.28.137.99	Broadcast
Wavence F Port #4 TMN Subnet	
Netmask: 255.255.255.252	
172.28.137.100	Network (/30)
172.28.137.101	**Wavence F Port #4**
172.28.137.102	F Ext Eqpt 1
172.28.137.103	Broadcast
Wavence E Port #4 TMN Subnet	
Netmask: 255.255.255.248	
172.28.137.104	Network (/29)
172.28.137.105	E Ext Eqpt 1
172.28.137.106	E Ext Eqpt 2
172.28.137.107	Spare
172.28.137.108	Spare
172.28.137.109	**Wavence E Port #4**
172.28.137.110	Router R2
172.28.137.111	Broadcast

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Wavence TMN Networking

TMN Network Planning – Growing the network

The unused 16 addresses from the original /26 allocation are split into subnets and assigned at sites G and H.

Wavence G TMN Ethernet Subnet	
Netmask: 255.255.252	
172.28.137.112	Network (/30)
172.28.137.113	**Wavence G TMN Eth (L)**
172.28.137.114	DHCP Assigned
172.28.137.115	Broadcast
Wavence H TMN Ethernet Subnet	
Netmask: 255.255.255.252	
172.28.137.116	Network (/30)
172.28.137.117	**Wavence H TMN Eth (L)**
172.28.137.118	DHCP Assigned
172.28.137.119	Broadcast
Wavence H Port #4 TMN Subnet	
Netmask: 255.255.255.248	
172.28.137.120	Network (/29)
172.28.137.121	**Wavence H Port #4**
172.28.137.122	H Ext Eqpt 1
172.28.137.123	H Ext Eqpt 2
172.28.137.124	H Ext Eqpt 3
172.28.137.125	Spare
172.28.137.126	Spare
172.28.137.127	Broadcast

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Wavence TMN Networking

TMN Network Planning – Growing the network

Wavence J TMN Ethernet Subnet	
Netmask: 255.255.255.248	
172.30.10.0	Network (/29)
172.30.10.1	DHCP Assigned
172.30.10.2	DHCP Assigned
172.30.10.3	DHCP Assigned
172.30.10.4	**Wavence J TMN Eth (L)**
172.30.10.5	J Ext Eqpt 1
172.30.10.6	Spare
172.30.10.7	Broadcast

Wavence K TMN Ethernet Subnet	
Netmask: 255.255.255.248	
172.30.10.8	Network (/29)
172.30.10.9	DHCP Assigned
172.30.10.10	DHCP Assigned
172.30.10.11	DHCP Assigned
172.30.10.12	**Wavence K TMN Eth (L)**
172.30.10.13	K Ext Eqpt 1
172.30.10.14	Spare
172.30.10.15	Broadcast

1. The new block of 16 addresses are assigned as shown.
2. OSPF will automatically handle route changes within the Wavence TMN Network.
3. External routers R1 and R2 will need additional static routes to use Wavence A and Wavence E respectively as gateways to reach the new 172.30.10.0/28 addresses.
4. With OSPF managing routes within the Wavence network and DHCP configuring the Craft computers the additional static routes added to the external routers are the only routing change required.

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B. Basics of IP Addressing

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IP Address Basics – Addressing standards

There are two types of IP addressing schemes:

1. **IPv4** - Internet Protocol version 4
    1. Most widely used addressing type

2. **IPv6** - Internet Protocol version 6
    1. Replacement for IPv4
    2. Authority : IANA - Internet Assigned Number Authority
    3. http://www.iana.org

Wavence TMN Management interfaces can be configured for either IPv4 (default) or IPv6. Simultaneous support for both IPv4 and IPv6 (dual stack) is not supported.

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IP Address Basics – IPv4 address

IPv4 address is a 32-bit binary number:

1010 1100 0001 0110 1000 1010 1100 1111

The most common representation uses dotted decimal notation such as:

172.22.138.207

Each of the four decimal numbers represents an octet or 8 bits of the 32-bit address. Each of the four numbers can range from 0 to 255. IPv4 provides an addressing capability of 2^{32} or about 4.3 billion addresses.

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IP Address Basics – IPv4 address parameters

The 32-bit IP addresses are divided into a Network prefix and a Host number. This particular example shows a 22 bits allocated for the network prefix and 10 bits for the host number:

172.22.138.207 -> 1010 1100 0001 0110 1000 10|10 1100 1111

network prefix | host number

There are two numbers reserved in each network, the first number and the last number. When the Host number portion of an IP address is all zeros it is called the Network Number. This is the first number in a Network:

1010 1100 0001 0110 1000 1000 | 0000 0000 -> 172.22.136.0

network prefix | host number

When the Host number portion of an IP address is all ones it is called the Broadcast Address. This is the last number in a Network:

1010 1100 0001 0110 1000 1011 | 1111 1111 -> 172.22.139.255

network prefix | host number

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IP Address Basics – IPv4 address parameters

The division between the network prefix and host number in an IP Address is specified using a Netmask. Like IP Addresses, Netmasks are 32- bit numbers and the usual representation is four dotted decimal numbers. Netmasks define the size or the number of hosts within a network.

Netmasks consists of:

1. a contiguous string of ones starting from the left end for the Network prefix portion
2. a contiguous string of zeros starting from the end of the Network prefix all the way to the right end for the Host number portion
3. No intervening bits

Using the example address from before with a 22- bit network prefix, the corresponding netmask is shown:

172.22.138.207 → 1010 1100 . 0001 0110 . 1000 1010 . 1100 1111

network prefix | host number

255.255.252.0 → 1111 1111 . 1111 1111 . 1111 1100 . 0000 0000

Table 1
Acceptable mask values

HEX	BIN	DEC
00	0000 0000	0
80	1000 0000	128
C0	1100 0000	192
E0	1110 0000	224
F0	1111 0000	240
F8	1111 1000	248
FC	1111 1100	252
FE	1111 1110	254
FF	1111 1111	255

Wavence supports recommendations in RFC1812 section 2.2.5.2: Net Mask Requirements for Classless Inter Domain Routing (CIDR) which allows the boundary between the network and host portions to be defined in 1- bit increments. The table to the right shows the allowed netmask values for an octet.

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IP Address Basics – Derivation of related network parameters

Netmask are utilized for ease in computation of related Network parameters.
A logical “AND” of the Netmask and Address gives the Network Number.
A logical “OR” of the Address with the *inverse* of the Netmask gives the Broadcast Address.
For example:
If my address were 10.100.49.30 and my netmask was 255.255.254.0 then I am a member of network 10.100.48.0, and my broadcast address is 10.100.49.255

```
10.100.49.30 ->  0000 1010 . 0110 0100 . 0011 0001 . 0001 1110 IP Address
255.255.254.0 -> 1111 1111 . 1111 1111 . 1111 1110 . 0000 0000 Netmask
                 ---------------------------------------------
  Logical AND    0000 1010 . 0110 0100 . 0011 0000 . 0000 0000 -> 10.100.48.0 Network

10.100.49.30 ->  0000 1010 . 0110 0100 . 0011 0001 . 0001 1110 IP Address
255.255.254.0 -> 0000 0000 . 0000 0000 . 0000 0001 . 1111 1111 Inverted Netmask
                 ---------------------------------------------
  Logical OR     0000 1010 . 0110 0100 . 0011 0001 . 1111 1111 -> 10.100.49.255 Broadcast
```

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Wavence TMN Networking

IP Address Basics – Describing networks (1/2)

There are two ways to describe networks:

1. **Long method:**
   Uses 3 numbers, at least two are required. The third number can be calculated.
   - Network Address (all 0's host)
   - Broadcast Address (all 1's host)
   - Netmask (leading 1's, trailing 0's)

2. **A shorthand method** also called CIDR (from RFC 4632):
   - In the form of the Network Number followed by a slash '/' and the Netmask length in decimal
   - Trailing zeros in the Network Number may be dropped.

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IP Address Basics – Describing networks (1/2)

CIDR Shorthand	Long method		Binary
Loop back Network			
127/8	NET	127.0.0.0	0111 1111.0000 0000.0000 0000.0000 0000
	MASK	255.0.0.0	1111 1111.0000 0000.0000 0000.0000 0000
	BCAST	127.255.255.255	0111 1111.1111 1111.1111 1111.1111 1111
One of the Reserved networks for private address space			
192.168/16	NET	192.168.0.0	1100 0000.1010 1000.0000 0000.0000 0000
	MASK	255.255.0.0	1111 1111.1111 1111.0000 0000.0000 0000
	BCAST	192.168.255.255	1100 0000.1010 1000.1111 1111.1111 1111
Sample Network Info			
143.209.100/22			
	NET	143.209.100.0	1000 1111.1101 0001.0110 0100.0000 0000
	MASK	255.255.252.0	1111 1111.1111 1111.1111 1100.0000 0000
	BCAST	143.209.103.255	1000 1111.1101 0001.0110 0111.1111 1111

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IP Address Basics – IPv4 Possible Network Sizes

Two hosts are reserved in any Ethernet Broadcast network for the Network Number, and the Broadcast address: the all 0's host and the all 1's host respectively. This means the number of useable hosts is always two less than the total number of addresses in the network.
The smallest Broadcast Ethernet network supported by Wavence is highlighted*.

Network bits	host bits	useable hosts	Decimal mask
31*	1	0	255.255.255.254
30	2	2	255.255.255.252
29	3	6	255.255.255.248
28	4	14	255.255.255.240
27	5	30	255.255.255.224
26	6	62	255.255.255.192
25	7	126	255.255.255.128
24	8	254	255.255.255.0
23	9	510	255.255.254.0
22	10	1022	255.255.252.0
21	11	2046	255.255.248.0
…	…	…	…

* Although RFC 3021 allows using a /31 network for point-to-point links Wavence does not support this extension.

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IP Address Basics – Subnet Calculators

Calculating network parameters can be difficult for those not familiar with the process. The are various online Network subnet calculators available that make derivation of all the related numbers relatively easy.

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IP Address Basics – IPv6 Addresses

IPv6 addresses are a 128 bits long. The most common representation uses eight groups of four hexadecimal words separated by colons (:)

2001:0db8:5160:c058:0000:8fd1:0513:fa0b

Each of the colon separated hexadecimal numbers represents one 16-bit word of the 128-bit address. Each of the eight words can range from 0000 to ffff (decimal: 0 to 65535). Addresses not case sensitive but using only lower case hex digits is recommended.

The binary representation of the above address is shown below:

```
0010 0000 0000 0001 0000 1101 1011 1000 0101 0001 0110 0000 1100 0000 0101 1000 0000 0000 0000 0000 1000 1111 1101 0001 0000 0101 0001 0011 1111 1010 0000 1011
 2    0    0    1 : 0    d    b    8 : 5    1    6    0 : c    0    5    8 : 0    0    0    0 : 8    f    d    1 : 0    5    1 : 3    f    a    0    b
```

The 128 bits provide for an address space of 2^{128} or about 3.4×10^{38} addresses. Most of the potential address space is unused by design. The longer addresses simplify the allocation of addresses, enable efficient route aggregation, and enable the implementation of special features like Stateless Auto Configuration.

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IP Address Basics – IPv6 Addresses simplification techniques

Due to the long length of IPv6 addresses there are methods to simplify written or displayed addresses. There are two basic rules:

1) Leading zeros may be removed from any hexadecimal word.
2) Consecutive words of zero may be replaced with a double colon (::). The double colon may only be used once in an address otherwise it becomes impossible to figure out how many zeros have been removed making the address ambiguous. RFC 5952 recommends that a double colon not be used to eliminate a single zero.

Consider the following IPv6 addresses:

2001:0db8:0000:000b:0000:0000:0000:001a
0000:0000:0000:0000:0000:0000:0000:0001 (IPv6 loopback address)

After applying rule 1)
2001:0db8:0000:000b:0000:0000:0000:001a becomes 2001:db8:0:b:0:0:0:1a
0000:0000:0000:0000:0000:0000:0000:0001 becomes 0:0:0:0:0:0:0:1

After applying rule 2)
2001:db8:0:b:0:0:0:1a becomes 2001:db8:0:b::1
0:0:0:0:0:0:0:1 becomes ::1

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IP Address Basics – Networks

As with IPv4 the 128-bit IPv6 addresses are divided into a Network prefix and a Host number using a Netmask.

IPv6 Netmasks use the CIDR (Classless Inter-Domain Routing) style shorthand notation of a slash (/) followed by a decimal number specifying the number of bits used for the network prefix.

For example, the network written as:

2001:db8:5160:c058::/64

indicates 64 bits allocated for the network prefix and defines a network with addresses in the range:

2001:db8:5160:c058:0000:0000:0000:0000
2001:db8:5160:c058:ffff:ffff:ffff:ffff

An address and netmask for a specific interface in the above network can be written as:

2001:db8:5160:c058::1/64

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IP Address Basics – Literal IPv6 address

When using IPv6 addresses with resource identifiers such as URLs the colon (:) conflicts with the established syntax of using a colon to designate a port number at the end of the host address. The workaround for this problem is to enclosed IPv6 addresses in square brackets [ ]

Web Browsers are probably the most often used application requiring square brackets with IPv6 addresses.

Example:

https://[fec0:0:0:1::1]/

If the URL requires a port number, it goes outside the square brackets:

https://[2001:db8:5160:c058:0:8fd1:513:fa0b]:80/

Square brackets are needed around an IPv6 address when connecting to the Wavence using a web browser.

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IP Address Basics – IPv6 alternative forms

The hexadecimal representation for IPv6 addresses is the most frequently used.

Some hybrid dual-stack IPv6/IPv4 implementations support IPv4-mapped IPv6 addresses. This is a transition technology to aid the migration to IPv6.

These addresses are typically written with a 96-bit prefix in standard IPv6 notation and the remaining 32 bits using a dotted decimal format:

::ffff:172.22.100.10

The Wavence does not support dual-stack or IPv4-mapped IPv6 addressing.

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IP Address Basics – IPv6 Multicast

IPv6 does not implement broadcast addressing.
The traditional role of a Broadcast address is replaced by Multicast addressing to the All-Nodes multicast group.
In general, it is not necessary to communicate with all nodes in a network so most IPv6 protocols that need to talk with multiple nodes use a dedicated multicast group to avoid disturbing all the devices in the network.
Multicast addresses generally take the form: ff0x::y
Common multicast addresses are:

- ff02::1 – All Nodes address
- ff02::2 – All Routers address
- ff02::5 – OSPFIGP
- ff02::6 – OSPFIGP Designated Routers

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Wavence TMN Networking

IP Address Basics – Wavence addressing

For IPv4

1. Private networks often use Addresses from those defined in RFC1918:
    1. 10.0.0.0/8
    2. 172.16.0.0/12
    3. 192.168.0.0/16
2. Globally unique addresses can be used if they are available.
3. To choose a globally unique address for the Local Address and use RFC1918 addresses for the TMN interfaces

For IPv6

1. Globally unique addresses can be used.
2. For private networks one suggestion would be to follow the recommendations in RFC4193 and choosing addresses from the fd00::/8 block (L bit set to 1) and selecting a suitable Locally Assigned Global ID.
3. To choose a globally unique address for the Local Address and use private addressing for the TMN interfaces.

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C. Communication in Networks

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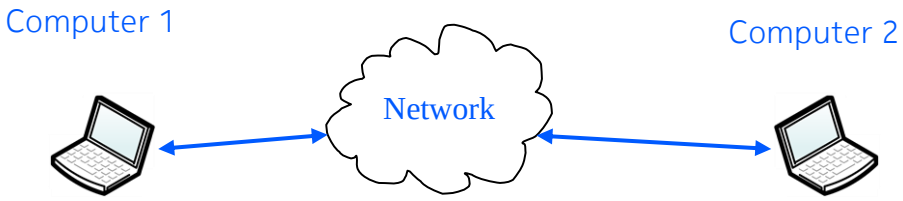
Wavence TMN Networking

Basic Networking concepts – Communication within a network



A simple local network using Ethernet to allow Computer 1 and Computer 2 to communicate with each other.
The network could be implemented with fiber, twisted-pair or some other technology.
The connection between the two computers could be direct using a single cable or with a switch.

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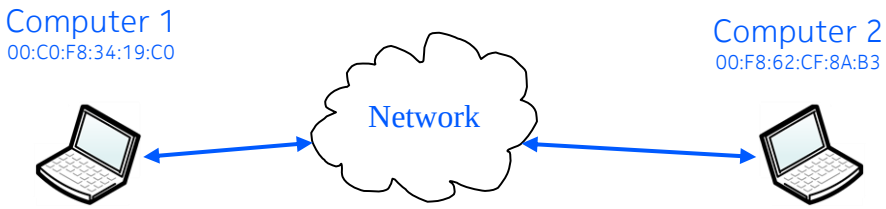
Wavence TMN Networking

Basic Networking concepts – Communication within a network



So how do two devices communicate using Ethernet?
A physical address is used to distinguish the two devices, this address is referred to as the MAC address. The **MAC address** is a 48-bit address assigned by the manufacturer of the network interface before it is shipped, it is intended to be unique, and is used to help identify a machine on a network.
MAC addressing is OK for direct Ethernet communication, but…

1. the end user usually has no control over the address
2. it is impractical outside a local network.

To make things easier for users, another communication layer is added: **IP Addressing**

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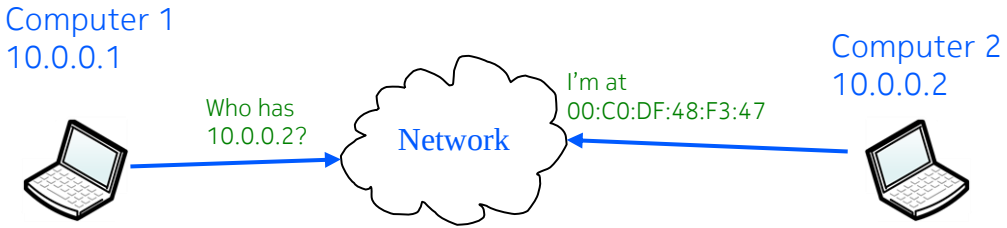
Wavence TMN Networking

Basic Networking concepts – Communication within a network



Even with IP Addressing, any time one device needs to talk with another using Ethernet, it still needs to know the MAC address for that device.

MAC addresses are resolved by queries on the local network asking the device you are trying to communicate with to send back its MAC address.

IPv4 uses the Address Resolution Protocol (ARP)
With IPv6 this is part of the Neighbor Discovery (ND) protocol.

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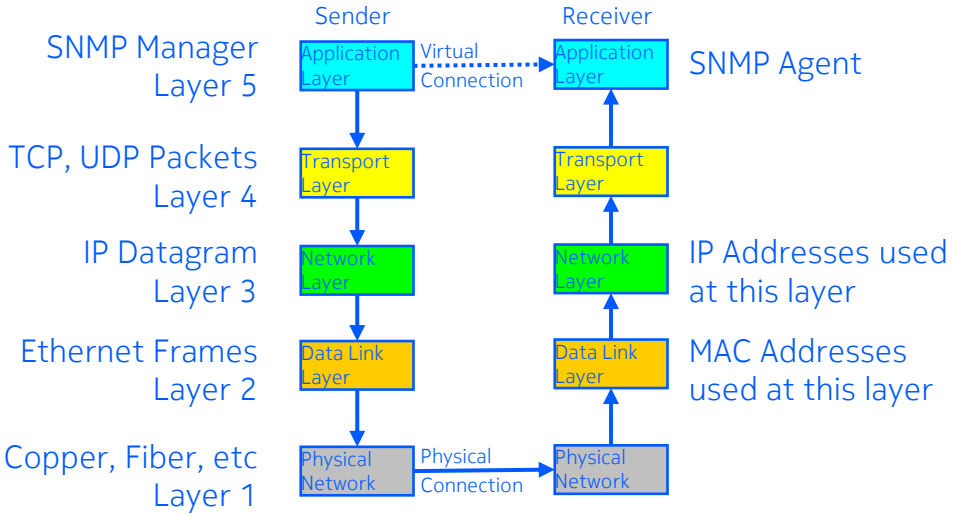
Wavence TMN Networking

Basic Networking concepts – TCP Stack

So how do the addressing methods fit together?



MAC addresses are used at **Layer 2**
IP Addresses are used at **Layer 3**

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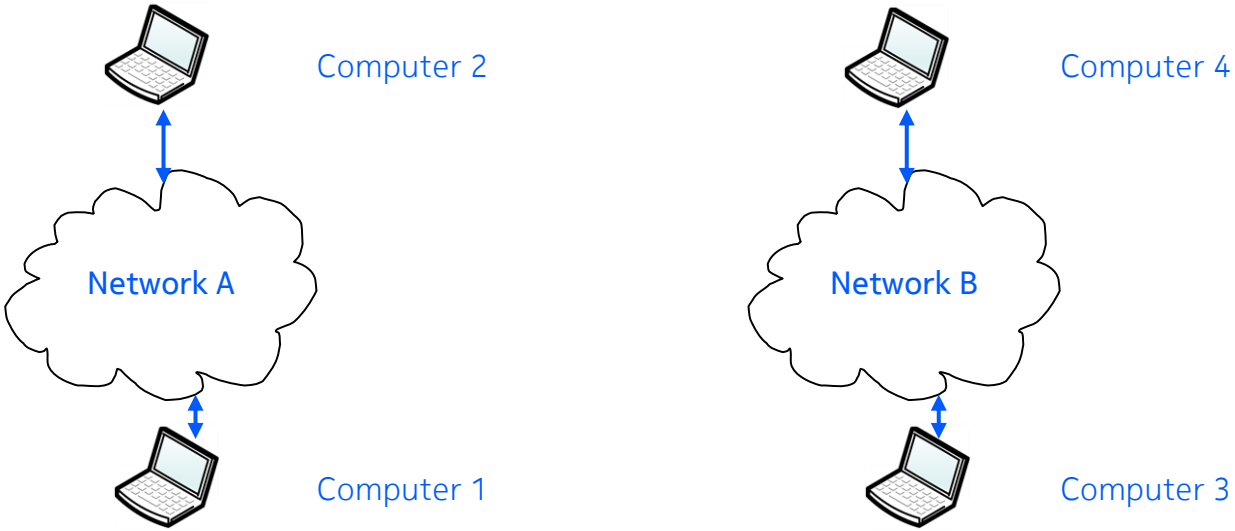
Wavence TMN Networking

Basic Networking concepts – Network communication



In this situation we have two isolated networks. Computer 1 can communicate with Computer 2, and Computer 3 can communicate with Computer 4.
But what if Computer 1 needs to send a message to Computer 4?
We need a way to interface the two Networks.

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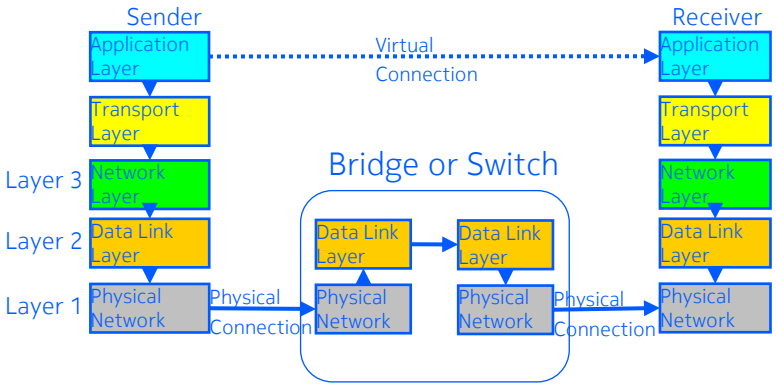
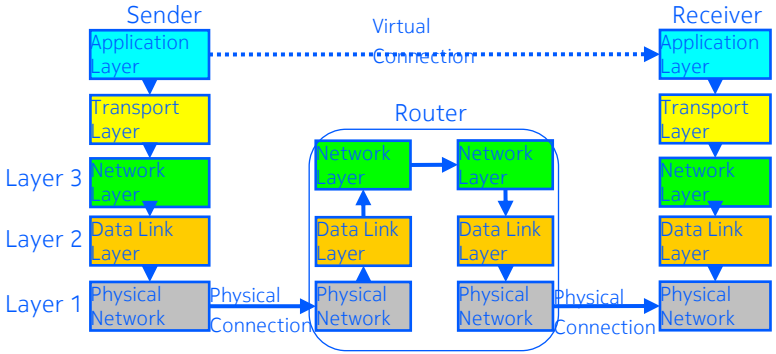
Wavence TMN Networking

Basic Networking concepts – Router and bridge comparison

A **bridge** or switch connects at Layer 2



Routers connect at Layer 3



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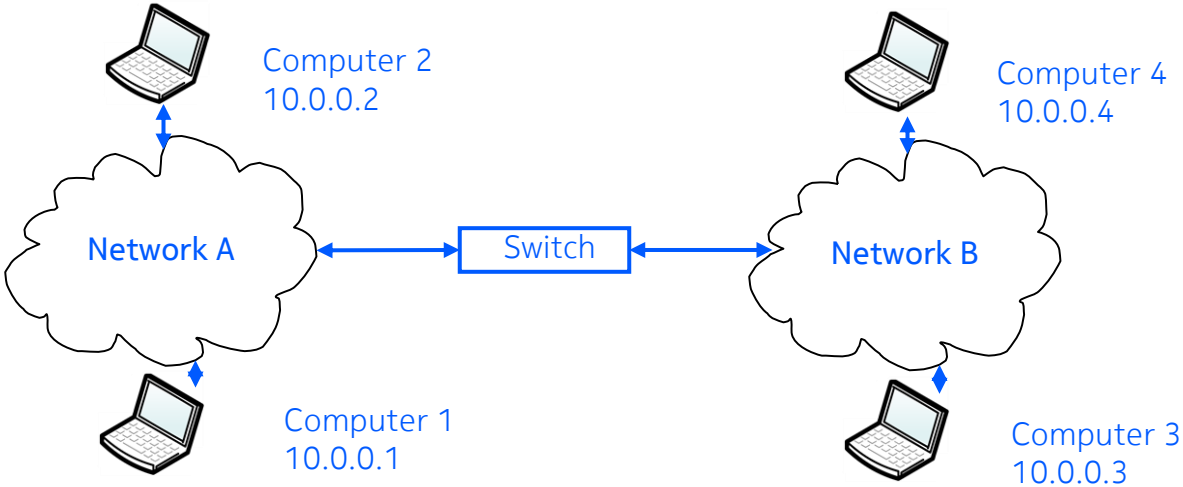
Wavence TMN Networking

Basic Networking concepts – Connecting networks



The common Ethernet Switch is a form of bridge. The interfaces on a bridge have no IP address. Bridges are convenient when all network devices share the same IP address space (the same subnet). Packets received at one port are essentially repeated on the other port(s). Most bridging devices are somewhat more intelligent about which packets they forward and where, but that's beyond the scope of this presentation.

Wavence TMN Network does not bridge between TMN Ports or across RF Links!

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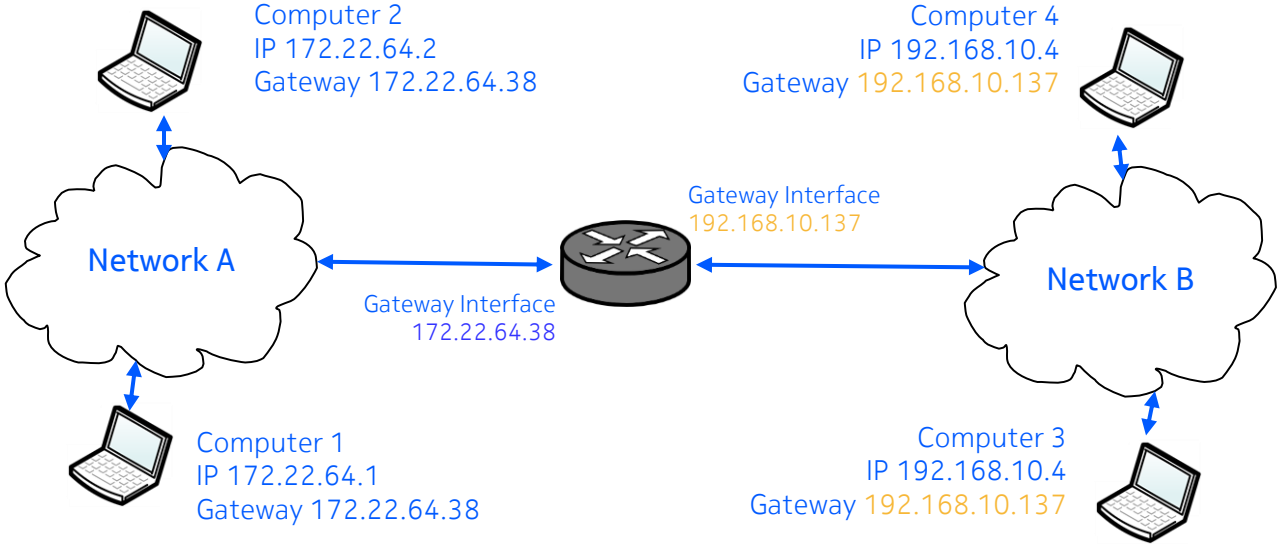
Wavence TMN Networking

Basic Networking concepts – Connecting networks



Routers are another way to connect two networks together. Routers are used when the two networks use different IP address space.

Unlike bridges and switches interfaces on a router need an address within the networks they are attached to. The interface IP address of a router is a ‘gateway’ to other networks. Most network devices are configured to use a nearby router as a Default Gateway.

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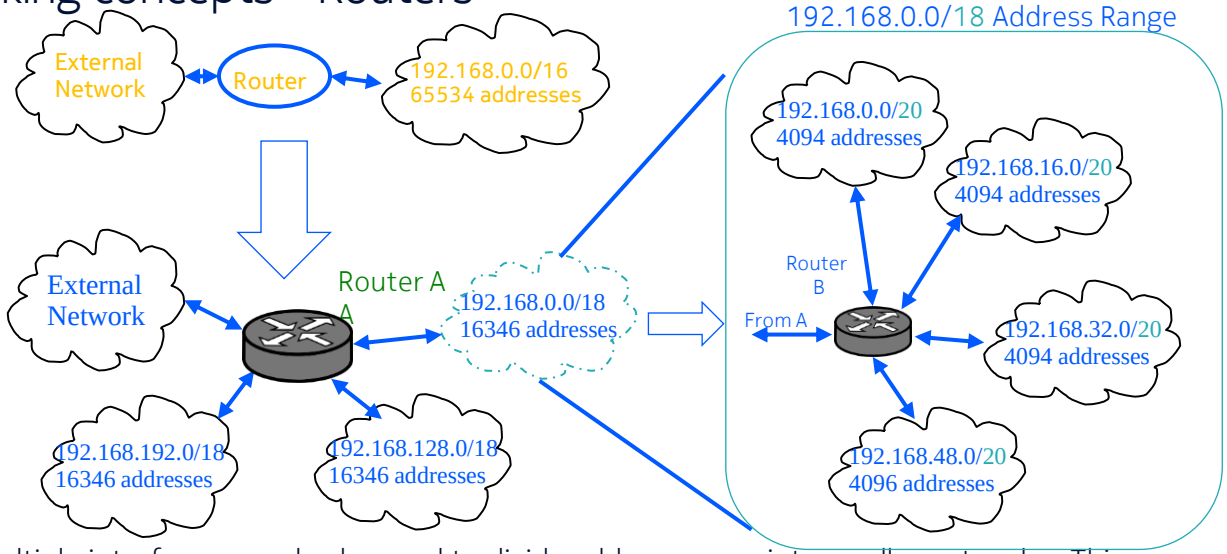
Wavence TMN Networking

Basic Networking concepts – Routers



1. A Router with multiple interfaces can also be used to divide address space into smaller networks. This process of division can be repeated to get the desired network size, optimizing use of the available address space. Each division is called a sub-network.
2. In the above example, the original 192.168.64.0/16 network at the top left is divided into four subnets as the length of the netmask is increased from 16 to 18 bits. Three of the new subnets are directly attached to Router A. One /18 subnet has been further divided into four /20 subnets using Router B as shown with the detail.
3. Router A doesn't need to know that Router B has divided the 192.168.64.0/18 network into subnets, it only needs to know that the 192.168.64.0/18 address space is accessible via Router B.

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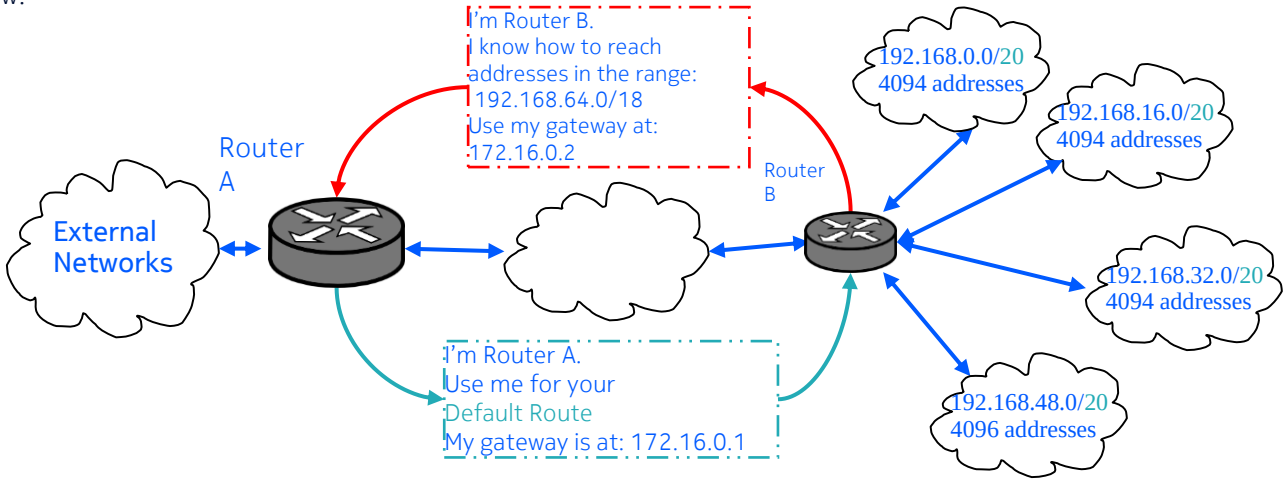
Wavence TMN Networking

Basic Networking concepts – Router configuration

In the Previous slide, Router A needs to know that addresses in the 192.168.64.0/18 network can be reached by using Router B as a gateway.

There are two methods considered:

1. Static Routing where all routes are manually provisioned.
2. Dynamic Routing where routers exchange route information using a Dynamic Routing Protocol. A Dynamic routing message exchange is depicted below:



There are several different dynamic routing protocols.

The dynamic routing protocol used by Wavence is called **OSPF (Open Shortest Path First)**. Refer to the OSPF Appendix for more information.

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D. Wavence DHCP Overview

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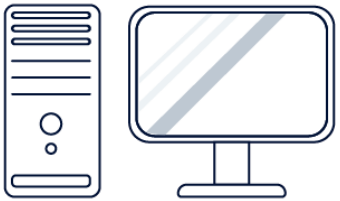
Wavence TMN Networking

Basic Networking Concept - DHCP

There are two address assignment methods:

1. Static assignment
2. Dynamic assignment

With static assignment, addresses are configured manually:



IP Address:
172.22.88.175
Netmask:
255.255.224.0
Default Gateway:
172.22.64.1

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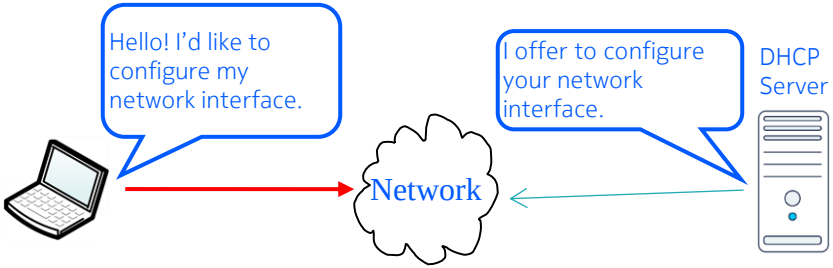
Wavence TMN Networking

Basic Networking Concept - Address Assignment for IPv4

With Dynamic Addressing equipment on a network is configured using the Dynamic Host Configuration Protocol (DHCP) as documented in RFC2131. Wavence provides a simple DCHP server on the TMN Ethernet port for configuring Craft computers. This DHCP server is enabled by Default.



With DCHP, when clients connect to a network, they send a broadcast asking if there is a server that can provide networking configuration.
If a DHCP Server is available, the client can then request its network configuration parameters.
When a DHCP Server provides an IP Address to a client it is called a Lease. The typical parameters provided usually include: IP Address, Netmask, Default Gateway, and Lease Timeout.

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Wavence TMN Networking

Basic Networking Concept - Address Assignment for IPv4

1. The DHCP Server is responsible for keeping track of which IP Addresses are currently Leased and not hand out duplicates.

2. When a client is finished with an IP address, it is supposed to inform the Server the address is no longer needed. The Server will then mark the address as available for reuse.

3. Leases have an associated timeout specified by the server. The Lease timeout is part of the configuration parameter message. This timeout is the maximum time the client is allowed to use the IP Address. Timeouts are specified in seconds and usually range from minutes to days.

4. If a client disconnects without informing the server or otherwise fails to renew the Lease then when the timeout arrives the Server will mark the address as available for reuse.

5. If a client needs an IP Address for an extended period, it must negotiate with the DHCP server to renew the Lease prior to timeout. If the Lease expires, the client must request a new Lease and may be assigned a different IP Address.

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Wavence TMN Networking

Basic Networking Concept - Address Assignment for IPv4

1. DHCP is based on the Bootstrap Protocol RFC951 (BOOTP)
    1. DHCP can be relayed between networks by a BOOTP Relay Agent. These are sometimes called DHCP Relay Agents.
    2. A BOOTP Relay allows DHCP clients and servers to be in different Networks. Some routers incorporate BOOTP Relay Agents.

2. Wavence TMN Network router does NOT Relay DHCP or BOOTP messages!

Multiple DHCP Servers on a Network:

1. This is possible provided either:
    1. The multiple DHCP Servers share a common Lease database so they do not serve duplicate addresses
    2. The Servers are configured to offer Leases from non-overlapping address ranges.
2. When a client broadcasts a request looking for a DHCP server and multiple servers respond the client chooses the server it will use for configuration. Frequently this is the first server to respond, but this behavior is not required.

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Wavence TMN Networking

Basic Networking Concept – DHCP in Wavence

1. Wavence DHCP service is only available with IPv4.
    1. When configured for IPv6 the Wavence provides link-local announcements using the Neighbor Discovery Protocol (NDP). This allows Craft computers to configure themselves through stateless auto-configuration.
2. Wavence DHCP server Leases addresses based on a very specific set of rules. These rules, defined on the following pages, are unique to Wavence and are not user configurable.
3. Wavence DHCP server defaults to Enabled and must be disabled before connecting the TMN Ethernet interface to an external network controlled by another DHCP server otherwise conflicts will occur including possible loss of connectivity in the external network.

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Wavence TMN Networking

Wavence DHCP Leased Address Assignment behaviour

1. For simplicity, the built-in DHCP server in Wavence operates using a fixed set of rules. This fixed behavior puts restrictions on the way addresses in the TMN Ethernet Port subnet can be used whenever the DHCP server is enabled.
2. The DHCP Server uses an address pool based on the TMN Ethernet IP address and subnet.
3. Up to 10 addresses are reserved for Leases in the following manner:
    1. If the TMN Ethernet Port address is the first address in the subnet, reserve addresses immediately above the Port to the end of the subnet or a maximum of 10 addresses, whichever comes first.
    2. If the TMN Ethernet Port address is NOT the first address of the subnet but there are 10 IP addresses above it before the end of the subnet reserve those 10 addresses for the DHCP pool.
    3. If the TMN Ethernet Port address is NOT the first address in the subnet, and there are fewer than 10 addresses above it before the end of the subnet then reserve addresses immediately prior to the port up to a maximum of 10 addresses or the start of the subnet whichever comes first.
4. If the Wavence DHCP Server is enabled, the addresses reserved for DHCP Leases should not be statically assigned to other equipment.
5. The manner in which DHCP addresses are reserved and assigned by the Wavence is shown with examples on the following slides.

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Wavence TMN Networking

Wavence DHCP Address Assignment behaviour

Table 1

IP Addr	Equipment
192.168.1.15	Broadcast
192.168.1.14	
192.168.1.13	DHCP Assigned 10
192.168.1.12	DHCP Assigned 9
192.168.1.11	DHCP Assigned 8
192.168.1.10	DHCP Assigned 7
192.168.1.9	DHCP Assigned 6
192.168.1.8	DHCP Assigned 5
192.168.1.7	DHCP Assigned 4
192.168.1.6	DHCP Assigned 3
192.168.1.5	DHCP Assigned 2
192.168.1.4	DHCP Assigned 1
192.168.1.3	**TMN Ethernet Port**
192.168.1.2	
192.168.1.1	
192.168.1.0	Network (/28)

Table 2

IP Addr	Equipment
192.168.1.15	Broadcast
192.168.1.14	
192.168.1.13	
192.168.1.12	**TMN Ethernet Port**
192.168.1.11	DHCP Assigned 10
192.168.1.10	DHCP Assigned 9
192.168.1.9	DHCP Assigned 8
192.168.1.8	DHCP Assigned 7
192.168.1.7	DHCP Assigned 6
192.168.1.6	DHCP Assigned 5
192.168.1.5	DHCP Assigned 4
192.168.1.4	DHCP Assigned 3
192.168.1.3	DHCP Assigned 2
192.168.1.2	DHCP Assigned 1
192.168.1.1	
192.168.1.0	Network (/28)

In Table 1, there are 10 addresses available above the TMN Ethernet Port address and they are reserved for the DHCP address pool (Rule 2b). Addresses not used by the TMN Ethernet Port or reserved for DHCP Leases can be statically assigned to other equipment.

In Table 2, there are fewer than 10 addresses available above the TMN Ethernet Port, but there are 10 available below it and they are reserved for the DHCP address pool (Rule 2c). Addresses not used by the TMN Ethernet Port or reserved for DHCP Leases can be statically assigned to other equipment.

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Wavence TMN Networking

Wavence DHCP Address Assignment behaviour

Table 3

IP Addr	Equipment
192.168.1.7	Broadcast
192.168.1.6	DHCP Assigned 5
192.168.1.5	DHCP Assigned 4
192.168.1.4	DHCP Assigned 3
192.168.1.3	DHCP Assigned 2
192.168.1.2	DHCP Assigned 1
192.168.1.1	**TMN Ethernet Port**
192.168.1.0	Network (/29)

Table 4

IP Addr	Equipment
192.168.1.7	Broadcast
192.168.1.6	
192.168.1.5	**TMN Ethernet Port**
192.168.1.4	DHCP Assigned 4
192.168.1.3	DHCP Assigned 3
192.168.1.2	DHCP Assigned 2
192.168.1.1	DHCP Assigned 1
192.168.1.0	Network (/29)

Table 5

IP Addr	Equipment
192.168.1.7	Broadcast
192.168.1.6	
192.168.1.5	
192.168.1.4	
192.168.1.3	
192.168.1.2	**TMN Ethernet Port**
192.168.1.1	DHCP Assigned 1
192.168.1.0	Network (/29)

In Table 3, the TMN Ethernet Port address is the first address in the subnet so addresses above the port will be reserved. Since there are fewer than 10 addresses to the end of the subnet all addresses are reserved for the DHCP address pool (Rule 2a). This leaves NO addresses available for static assignment to other equipment.

In Tables 4 and 5, the TMN Ethernet Port address is not the first address in the subnet and there are fewer than 10 addresses available above the port so addresses below the port will be reserved. With fewer than 10 addresses available below the port all are reserved for the DHCP address pool (Rule 2c). Addresses not used by the TMN Ethernet Port or reserved for DHCP Leases may be statically assigned to other equipment.

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E. Wavence OSPF Overview

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Wavence TMN Networking

Network services – OSPF Overview

1. Open Shortest Path First (OSPF) is the dynamic routing protocol used with Wavence.

2. In IPv4 mode Wavence uses OSPFv2 as defined in RFC 2328.

3. In IPv6 mode Wavence uses OSPFv3 as defined in RFC 5340.

4. OSPF is a link-state protocol. We can think of a link as an interface to a router, and the associated link state as a description of that interface and it's relation to other routers. The link state includes such info as:
   1. IP Address of the interface
   2. Netmask
   3. The type of network
   4. The routers connected to that interface

5. The collection of these link-states for several interfaces would form a link-state database

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Wavence TMN Networking

Network services – OSPF Overview

1. Routers running OSPF advertise their link state to neighboring routers at initialization and again whenever any link state information changes. The advertisement represents the collection of all link states on that router.

2. Routers exchange link states by means of flooding. Whenever a router receives a link state update, it stores a copy in it's local database and propagates the update to other routers.

3. After the database is updated, the router will calculate the Shortest Path tree to all destinations. The destinations, the cost, and the next hop to reach those destinations form the IP routing table.

4. To control the flooding of link state updates, OSPF uses Areas. All routers within an Area have the same link state database.

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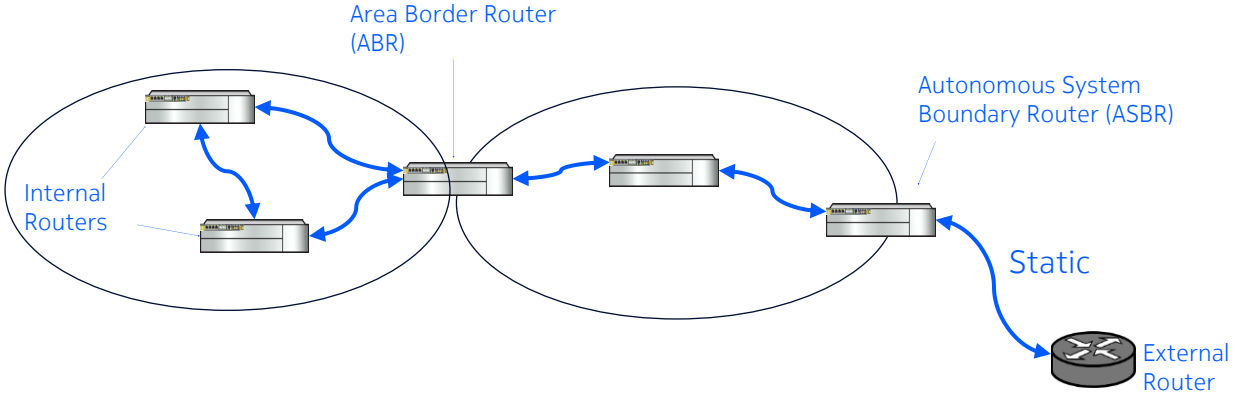
Wavence TMN Networking

Network services – OSPF Overview



1. Routers that have all of their interfaces in the same area are called Internal Routers.

2. Routers that belong to one or more areas and connect to the backbone must keep a link state database for all attached areas plus the Backbone Area. These routers are called Area Border Routers (ABR)

3. A router that act as a gateway between OSPF and another routing protocol (including Static routes) is called an Autonomous System Boundary Router (ASBR).

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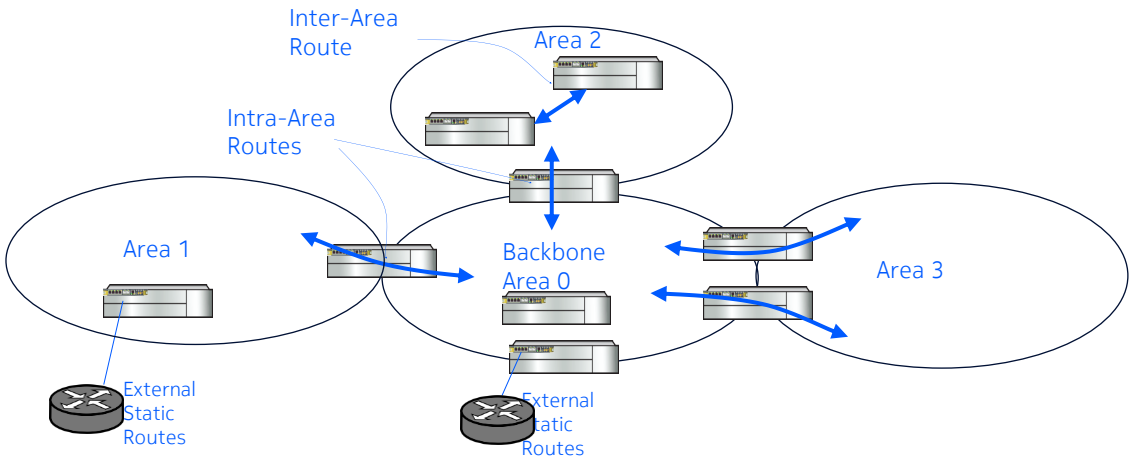
Wavence TMN Networking

Network services – OSPF Overview



OSPF has special restrictions when multiple Areas are involved:

1. If more than one area is configured, ONE of these areas must be Area 0. This area is defined as the Backbone. When designing networks, it's good practice to start with Area 0 and expand into other areas later on.
2. The backbone has to be at the center of all other areas. All other areas must be physically attached to the backbone. OSPF expects all non-zero areas to inject routes into the backbone and the backbone will redistribute this information to the other areas.

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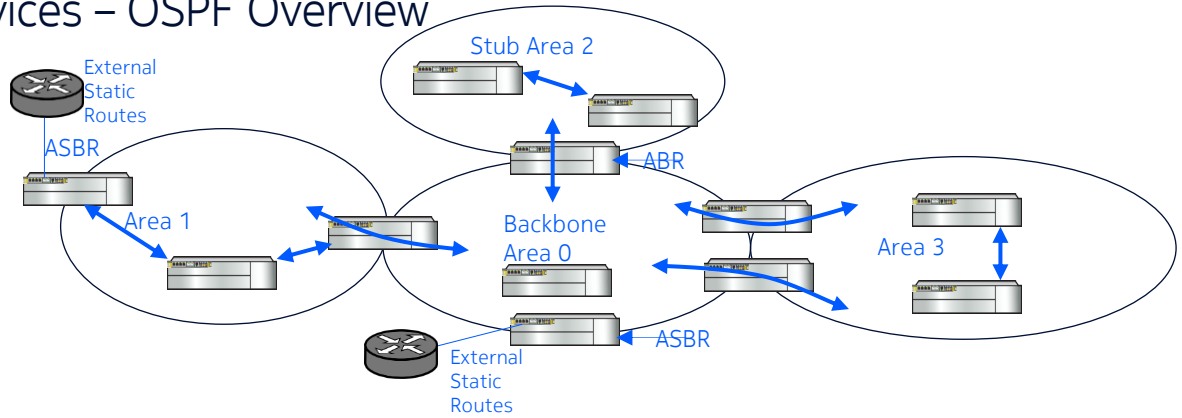
Wavence TMN Networking

Network services – OSPF Overview



Stub Areas:

1. OSPF allows certain areas to be defined as Stub Areas. Routes to external networks, including those learned through other protocols are not allowed to be flooded into these areas. Routing from Stub areas to the outside world is based on a default route to the nearest ABR. Configuring an area as a Stub reduces the size of the routing database within the area.
2. An area could be qualified as a Stub if there is a single exit point from the area or if routing to destinations outside the area do not need to take an optimal path. In the example above, Area 2 is shown as a Stub, Area 3 could be a candidate for a Stub.
3. Stub areas can have more than one ABR but must not contain any ASBR.
4. All routers within a Stub area must have the Stub Flag set.

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Wavence TMN Networking

Network services – OSPF Overview

Wavence OSPF Features

1. Each Wavence supports up to 3 non-zero user defined OSPF Areas, for a total of 4 Areas.
2. Area 0 is always pre-defined in each Wavence and cannot be deleted.
3. The Wavence automatically becomes an Area Border Router (ABR) when there are active interfaces assigned to different OSPF Areas. ABRs must have at least one active interface associated with Area 0.
4. Wavence will act as an Autonomous System Border Router (ASBR) when:
    1. OSPF is selected as the Routing Protocol for a TMN Interface
    2. At least one Static Route is configured to reach a Gateway through that TMN Interface
    3. There is no OSPF Adjacency with the Gateway router
5. Stub Area support
    1. Set the Stub Flag to True on each Wavence in the area. The Wavence can only be configured for Stub True or False. There is no support for Not So Stubby Areas (NSSA)
    2. The Wavence has no options to suppress Type 3 Summary LSAs (Totally Stubby) or to configure route summarization. To limit the size of the route table in a Stub area use a more configurable external router for the ABR and restrict the routes advertised into Wavence OSPF Areas or configure the interface on the external ABR for Totally Stubby operation.
6. Authentication is not supported.

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Wavence TMN Networking

Network services – OSPF Overview

Wavence OSPF Features cont.

1. RouterID (RID)
    1. With OSPFv2 the Local Address is automatically used as the RID.
    2. With OSPFv3 the RID must be explicitly set and must be unique within the autonomous system.

2. Router Priority fixed to 1.
    1. Router Priority affects the Designated Router (DR) Election process.
    2. If you want external routers to have a greater chance of becoming the DR configure them to have a Router Priority greater than 1.
    3. Note that if the DR goes down or becomes unreachable Wavence might become the DR. If it is possible for a Wavence device to become the DR be aware the maximum number of direct neighbors supported in the same Broadcast network is eight.

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Wavence TMN Networking

Network services – OSPF Overview

1. Wavence OSPF Deployment Hints

2. The recommended limit for the number of entries in the Wavence routing table, including both Static and Dynamic entries, is around 200.

3. Route tables with more than 250 entries are not supported. This provides sufficient margin for OSPF convergence and proper operation. When the maximums are reached or exceeded random routes will be silently dropped with a consequent impact on remote management. Dropped routes do not raise any faults or alarms. If the route to the external gateway is dropped all management communication with the Wavence network can be lost.

4. To estimate the size of the OSPF routing table in an Area sum together:
    1. 1 entry for each active RF-PPP link
    2. 1 entry for each active TMN In-Band interface in each Wavence (up to two)
    3. 1 entry for each active TMN Port #4 interface in each Wavence
    4. 1 entry for each active Local TMN Ethernet Interface in each Wavence
    5. 1 entry for the NE Local Address (only if it does not match an interface address)
    6. 1 entry for each static route provisioned in the Wavence network.
    7. 1 entry for each route distributed into the Area from the ABR

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Wavence TMN Networking

Network services – OSPF Overview

Wavence OSPF Deployment Hints cont.

1. Example: A typical repeater supporting:
    1. two Radio Directions
    2. an active TMN Port #4 interface with a matching Local Address
    3. an active TMN Ethernet port
2. will contribute up to 4 route entries to the OSPF routing tables.
3. The upper limit is around 50 Wavences in an autonomous OSPF system.
    1. This is based on a target maximum route table size of about 200 entries and an average contribution of about 4 active route entries per Wavence: (200 / 4 = 50)
    2. Note: Peak TMN traffic loads caused by MIB Backups or Software Download can lead to excessive congestion in the rate limited TMN channel. To limit congestion the practical number of nodes in an autonomous Wavence OSPF system accessed through a single gateway is usually less than 30. Refer to the Deployment Hints later in this section.
4. If a Wavence OSPF area is attached to an external OSPF system it may be necessary to provision the external router to act as the ABR and restrict Type 3 Summary LSAs flooded into the Wavence area or else use route summarization to keep the size of the route table within recommended limits. One way to achieve this would be to configure the external router interface to the Area for Total Stub (stub, no-summaries).

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Wavence TMN Networking

Network services – OSPF fixed parameter

It may be necessary to configure several OSPF parameters on external routers to match the fixed values in Wavence before an adjacency can be formed.

The fixed Wavence OSPF parameters are:

1. Hello Interval: 10 seconds
2. Router Dead Interval: 40 seconds
3. MTU: 1500 bytes
4. Retransmit Interval: 5 seconds
5. Interface Transit Delay: 1 second
6. Router Priority: Fixed to 1 (lowest priority)
7. Note: In an IPv6 environment, the maximum size LSA supported by Wavence is 1280 bytes. LSAs larger than this will be discarded.

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Wavence TMN Networking

Network services – OSPF Deployments Hints

Other Wavence OSPF Deployment Hints cont.

1. Wavence will not form Full adjacency with a neighboring OSPF router if:
    1. The Area IDs do not match
    2. There is a mismatch in Hello Interval, Dead Time, or MTU size
    3. The neighboring router is configured to require authentication.
    4. or other incompatible settings.

2. changing OSPF areas on an interface can lead to loss of remote connectivity.
    1. **Caution:** Once OSPF Adjacency has been established any changes to the OSPF Area Id associated with the Wavence TMN interface will cause a loss of Adjacency through that interface. Changing the Area Id of the opposite end of the link to match will restore Adjacency but if one of the routers happens to be an ABR an NE Restart may be required at both ends of the link before routes will be exchanged correctly. This may lead to loss of remote connectivity and require a site visit to restart the Wavence and correct any other provisioning issues.
    2. An NE Restart can be performed using the ‘Supervision ->Restart NE’ function in the Craft.

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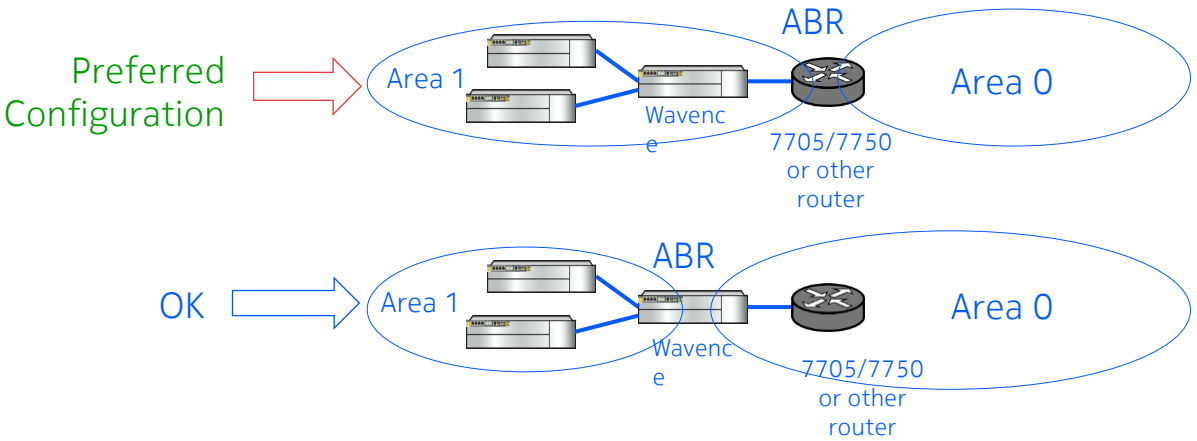
Wavence TMN Networking

Network services – OSPF Deployments Hints

Attaching a Wavence OSPF Area to an external OSPF network.

1. When the Wavence forms an adjacency with a more fully featured external router (such as an Nokia 7705 SAR, 7750, or other) at an OSPF Area border the best results will be obtained when the external router is configured as the ABR.



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Wavence TMN Networking

Network services – OSPF Deployments Hints

Other Wavence OSPF Deployment Hints
Autonomous Wavence OSPF systems with more than one ASBR may exhibit external reachability issues in the event of a link failure.

Workarounds:

- Use more fully featured external routers for the ASBRs.
- Integrate the Wavence OSPF system with an external OSPF system.
- Configure the Wavence OSPF area as a Stub to help limit the number of routes advertised into the area.
- In some cases it may be necessary to use an external ABR and configure the interface to the Wavence for Totally Stubby operation (stub with no summaries).

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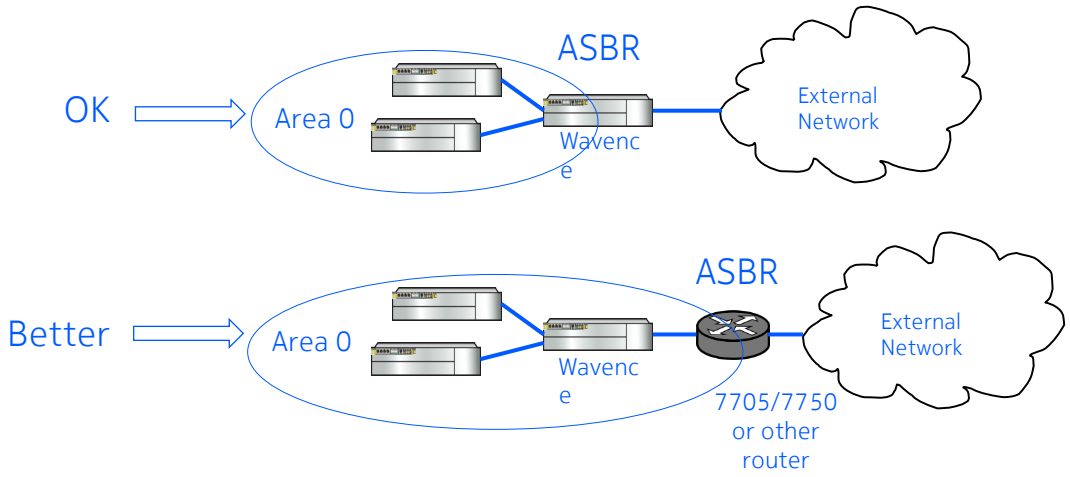
Wavence TMN Networking

Network services – OSPF Deployments Hints

Connecting a Wavence Autonomous OSPF network to external networks.

1. Autonomous networks will normally require a static route in the ASBR.
2. If a more fully featured external router is available (such as a Nokia 7705 or 7750 or other) using the external router as the ASBR will improve the performance.
3. If there will be multiple ASBRs in the network the external router solution is preferred.



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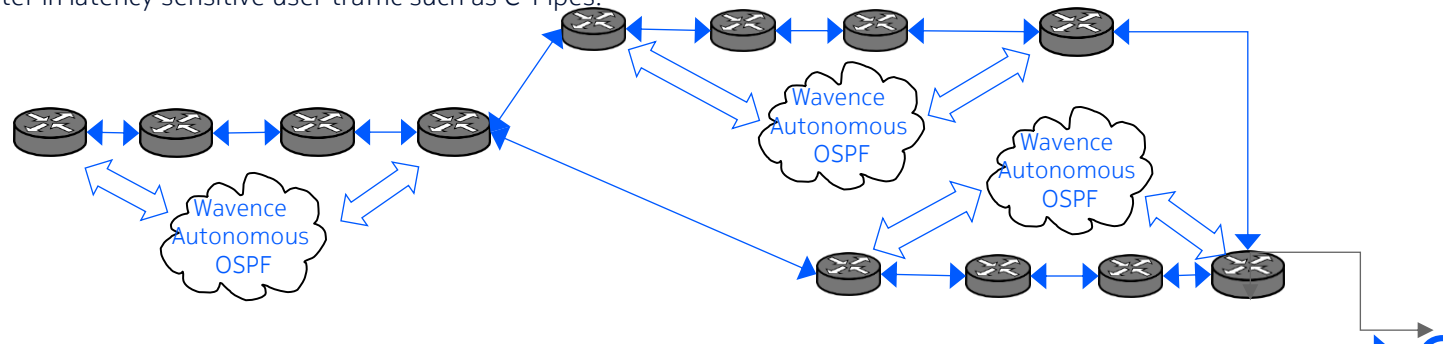
Wavence TMN Networking

Network services – OSPF Deployments Hints

Option 1 – Deploying Wavence in Large Networks

1. Configure Wavence into several autonomous OSPF networks using TMN RF-PPP links. Interface with the external Network as needed to limit the size of each Wavence OSPF network. Configure the interfaces in the external routers as an ASBR.
    1. Run an alternate routing protocol (like ISIS) in the external network. Distribute routes learned from Wavence OSPF into the external routing protocol. The ASBRs should Inject only a default route into each Wavence OSPF area.
2. Advantages
    1. Field maintenance of the Radio network does not depend on availability of the external routed network
3. Disadvantages
    1. Possibility of congestion in the RF-PPP links if there are too many Wavence devices behind an ASBR gateway – Divide the autonomous Wavence OSPF network into smaller regions.
    2. When used with low bitrate radios (25Mbit) bursts of TMN RF-PPP traffic during Wavence SW download or MIB Backups can cause excessive jitter in latency sensitive user traffic such as C-Pipes.



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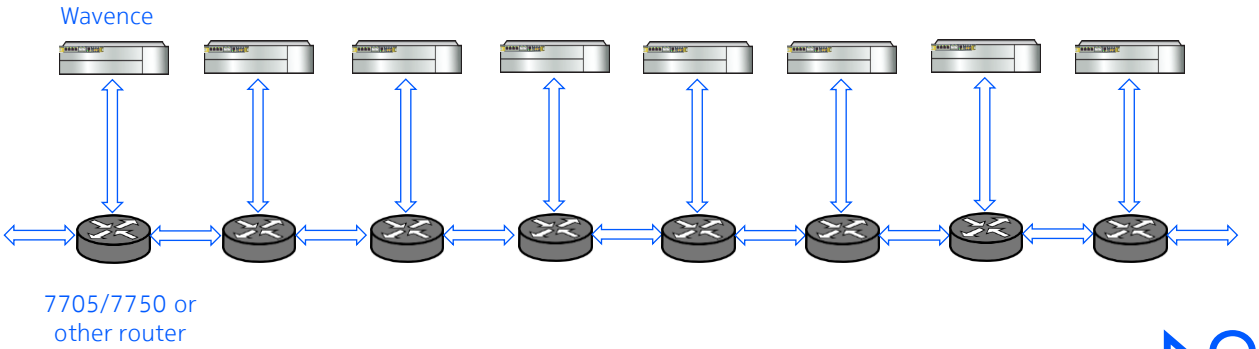
Wavence TMN Networking

Network services – OSPF Deployments Hints

Option 2 - Large Networks and with external routers available at each site

1. Manage each Wavence as a local device connected to the external router using either TMN Port #4 or a TMN In-Band VLAN port with a static default route pointing to the local external router.
2. Advantages
    1. Static routing avoids Wavence OSPF limitations
    2. The external routers have complete control of QoS for the Wavence TMN traffic.
3. Disadvantages
    1. Technician craft access across the hop to the far-end radio may only be available when the external routed network is operational. Enabling OSPF on RF-PPP links may provide a workaround as long as you stay within the Wavence OSPF limitation but it can also introduce additional failure modes.



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Wavence TMN Networking

Primary difference in IPv6 vs IPv4 in TMN

The major differences when running IPv6 are:

1. No DHCP support for the TMN Ethernet port when in IPv6 mode. Instead, the Wavence uses the IPv6 Stateless Address Auto Configuration (SLAAC). As a side benefit, all TMN ports support SLAAC.
2. In IPv4 mode, alarming of the TMN Ethernet port was automatically suppressed whenever DHCP was enabled. When using IPv6 it is necessary to manually provision suppression of the alarm if the port will not normally be connected. Alarm suppression is configured by changing the Alarm Profile on the TMN Ethernet port provisioning screen to: No Alarms.
3. IPv6 uses OSPFv3. With IPv4 and OSPFv2 the OSPF Router ID was automatically set to match the Wavence Local Address. With IPv6 this is no longer possible. The OSPFv3 Router IDs are still 32 bits long, but IPv6 addresses are 128 bits. When using OSPFv3 the Router ID must be manually set to a unique value in the overall OSPF system.
4. No dual stack operation. Wavence TMN management will be either all IPv4 or all IPv6. There are no settings associated with User traffic. Both IPv4 and IPv6 can be carried simultaneously.

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